

EGFR Variant Detection Panel Validation

June 4, 2026

Abstract

This report presents a bioinformatic validation of a targeted amplicon sequencing panel designed to detect clinically actionable mutations in the *EGFR* gene (epidermal growth factor receptor) relevant to non-small-cell lung cancer (NSCLC). The panel covers four amplicons spanning the key mutation hotspots in exons 18–21, including the exon 19 in-frame deletions (e.g. E746_A750del) and the exon 21 L858R substitution that confer sensitivity to tyrosine kinase inhibitors (TKIs) such as gefitinib, erlotinib, and osimertinib.

Using a single paired-end Illumina sequencing run (pre-aligned BAM, hg19), the analysis addresses five validation questions: (1) read constitution and adaptor content, (2) target gene identification by BLAST-based primer alignment, (3) overall mapping rate and read quality, (4) per-amplicon coverage depth and uniformity, and (5) somatic variant calling with orthogonal callers (Mutect2, VarDictJava). The analysis was implemented as a reproducible Snakemake pipeline running inside a Docker container, with all results compiled into this report.

Key findings: all four amplicons amplify on-target EGFR exon sequences; mapping rates exceed 99%; per-amplicon depth is uniformly high ($>1,000\times$) for three amplicons, with Amplicon 2 showing substantially lower coverage ($\sim 9,000$ vs. $\sim 200,000$ read pairs) attributable to primer inefficiency. Furthermore, 89% of Amplicon 2 read pairs are unmapped: BLAST analysis of a random subsample of 200 R1 reads (saved as `amp2_unmapped_sample.fa`) identifies 87% as originating from bacteriophage ϕ X174 (*Sinheimervirus phiX174*), the standard Illumina sequencing spike-in, indicating that the Amplicon 2 forward primer cross-reacts with the phage genome during PCR. Both findings suggest that the Amplicon 2 primer pair warrants re-design. One confirmed somatic EGFR driver mutation was identified: the exon 19 in-frame deletion E746_A750del (VAF 48%, VarDictJava, COSM6223), annotated by SnpEff as `p.Glu746_Ala750del` using the hg19 database. T790M and L858R are absent, confirmed by high-depth coverage ($>200,000\times$) at both hotspot positions with no PASS calls. A germline SNP at chr7:55,249,063 (rs1050171, EGFR Gln787Gln, synonymous) was flagged by both callers in tumour-only mode, illustrating the need for population germline filtering or a matched normal in clinical workflows. Two additional Mutect2 PASS calls at amplicon boundaries are primer-end artefacts caused by the absence of primer trimming (`ivar trim`) in the original BAM processing; these must be filtered before clinical reporting.

Contents

1	Biological Background: EGFR and Lung Cancer	4
2	Background: Amplicon Sequencing	7
2.1	Workflow	7
2.2	Read Structure	8
3	Dataset Description	9
3.1	Provided Files	9
3.2	Primer and Adaptor Sequences	9
3.3	BAM Header Analysis	9
4	Read Quality and Constitution	10
4.1	Raw Read Quality Assessment	11
4.1.1	Methods	11
4.1.2	Results	11
4.2	Read Constitution	13
4.2.1	Methods	13
4.2.2	Adaptor verification: which adaptor appears in each read?	14
4.2.3	Primer verification	16
4.2.4	Per-amplicon insert size distribution	17
4.2.5	Summary: read constitution	18
5	Gene Region Mapping	19
5.1	Primer Localisation by BLAST	19
5.2	Coverage-Based Region Identification	26
6	Mapping Rate and Unmapped Reads	28
6.1	Methods	28
6.2	Results	28
6.2.1	Which amplicons do the unmapped reads target?	29
6.2.2	Why are 89% of Amplicon 2 read pairs unmapped?	30
7	Amplicon Coverage	32
7.1	Methods	32
7.2	Results	33
8	Variant and Mutation Identification	36
8.1	Methods	37
8.2	Results	39
A	Confirming Primer-End Artefacts	43
B	How PCR Primers Work	44
C	Adaptor Scenario Verification	44
D	Per-amplicon Read-through Statistics	45
E	Implications of Amplicons Being Longer Than the Read Length	46

1 Biological Background: EGFR and Lung Cancer

Clinical use of EGFR mutation testing in lung cancer. Non-small-cell lung cancer (NSCLC) accounts for approximately 85% of all lung cancer cases and carries a poor prognosis when diagnosed at advanced stage. In roughly 10–15% of Western and 30–40% of East Asian NSCLC patients, the tumour is driven by an activating mutation in the *EGFR* gene. These patients are identified at diagnosis by targeted sequencing of the EGFR kinase domain (exons 18–21), which is now a mandatory step before initiating systemic therapy. A positive result directly determines the treatment choice: EGFR-mutant patients receive an oral EGFR tyrosine kinase inhibitor (TKI)¹ rather than conventional chemotherapy, with substantially better response rates and tolerability [10].

Tyrosine kinase inhibitor generations. A **sensitising mutation** is one that makes the tumour cells vulnerable to a TKI: the mutation keeps EGFR permanently active, and the drug binds to exactly that active conformation to switch it off. Without the mutation, the drug has no specific target and normal EGFR function would be disrupted. E746_A750del, L858R, and G719X are classic sensitising mutations; patients whose tumours carry them respond strongly to TKIs, whereas those without an EGFR sensitising mutation do not benefit.

Three generations of EGFR-targeted TKIs have been developed, each designed to overcome limitations of the previous one:

- **First generation** (gefitinib, erlotinib): the original EGFR inhibitors, in clinical use since 2003–2004. Effective against the common sensitising mutations but overcome by resistance in most patients within 12–18 months, primarily through acquisition of T790M.
- **Second generation** (afatinib, dacomitinib): approved from 2013 onwards; more potent and active against a broader range of mutations including G719X, but still overcome by T790M.
- **Third generation** (osimertinib): approved in 2015 initially for T790M-positive patients who had progressed on a prior TKI, and in 2018 as first-line therapy for all sensitising mutations due to superior progression-free survival and CNS penetration.

The clinical consequence is that the sequencing result not only confirms whether the patient is EGFR-mutant, but also which generation of TKI is appropriate: a treatment-naïve patient with L858R receives osimertinib (or a 1st/2nd-gen agent); a patient who has progressed on a 1st-gen TKI and newly acquired T790M is switched to osimertinib. Figure 1 summarises the decision logic.

¹A tyrosine kinase inhibitor is a small molecule drug that blocks the enzymatic activity of a kinase — a protein that switches other proteins on or off by attaching phosphate groups to them. EGFR is a receptor tyrosine kinase: when mutated, it is permanently switched on, driving uncontrolled cell growth. A TKI fits into the active site of the kinase and blocks this signal, effectively turning the driver mutation off.

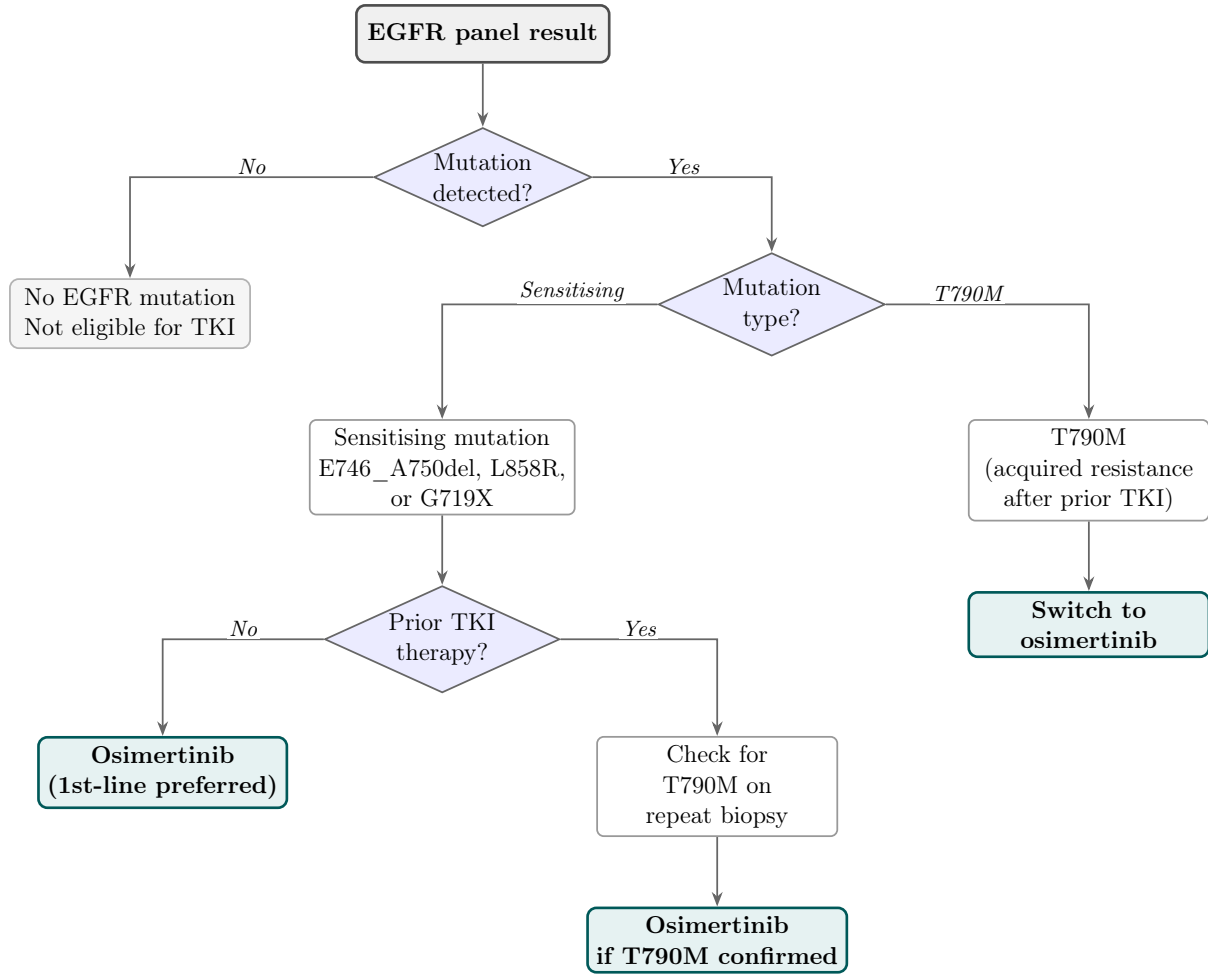


Figure 1: Treatment decision tree based on EGFR panel result. Sensitising mutations (E746_A750del, L858R, G719X) in treatment-naïve patients lead directly to osimertinib [14]. Patients who progress on a prior TKI are re-tested for T790M; a confirmed T790M triggers a switch to osimertinib [12].

Mutations targeted by this panel. The four amplicons in this panel each target one exon of the EGFR kinase domain, collectively covering the four most clinically relevant mutation classes (Table 1).

Table 1: Clinically relevant EGFR mutations targeted by this panel, with their exon location, covering amplicon, and clinical role in NSCLC. Freq.: percentage of EGFR-mutant NSCLC patients carrying each mutation [10]; the T790M figure applies to patients who have progressed on a prior TKI, as it is an acquired resistance mutation not present at diagnosis. Cross-reference with Table 25 for detected variants.

Mutation	Exon	Amplicon	Freq.	Clinical role
G719X	18	2	~3%	Sensitising; responds to afatinib and osimertinib
E746_A750del	19	1	~45%	Most common sensitising mutation; responds to all TKI generations
T790M	20	4	~50%	Acquired resistance to 1st/2nd-gen TKIs; sensitises to osimertinib
L858R	21	3	~40%	Second most common sensitising mutation; responds to all TKI generations

EGFR-mutant NSCLC cell lines. The four EGFR hotspot mutations are each well-represented by established non-small-cell lung cancer cell line models, which are widely used for drug-sensitivity testing and as sequencing controls (Table 2). Each line carries a defined EGFR genotype, making them useful reference standards: a sequencing run on a known cell line mix can be used to validate that a panel detects the expected mutations at the expected allele frequencies.

Table 2: EGFR mutation status of commonly used NSCLC cell lines for the four hotspots targeted by this panel. ✓ = mutation present; — = wild-type at that position. The profile of our sample (E746_A750del only, T790M/L858R absent) matches PC9 and HCC827.

Cell line	G719X	E746_A750del	T790M	L858R
PC9	—	✓	—	—
HCC827	—	✓	—	—
H1975	—	—	✓	✓
H3255	—	—	—	✓
NCI-H1650	—	✓	—	—
A549	—	—	—	—

A549 is EGFR wild-type (KRAS G12S) and serves as a TKI-insensitive negative control. H1975 is the canonical model for acquired T790M resistance, as it expresses both the primary sensitising mutation (L858R) and the resistance mutation (T790M) simultaneously. PC9 and HCC827 both carry the most common exon 19 deletion (E746_A750del) and are highly sensitive to all generations of EGFR TKIs.

Canonical genomic coordinates (hg19). Table 3 lists the precise reference coordinates for each mutation class. The nucleotide changes are given in VCF convention (plus-strand reference allele first), consistent with the output of Mutect2 and VarDictJava. These coordinates allow manual verification: querying the relevant position in

UCSC Genome Browser (hg19 assembly) or IGV should show the expected reference base, and the variant can be confirmed by inspecting the aligned reads in `aln.bam`.

Table 3: Canonical hg19 genomic coordinates of the EGFR mutations targeted by this panel, sourced from COSMIC [15] and ClinVar [8]. Ref/Alt follow VCF convention (plus-strand reference). G719X encompasses several codon-719 substitutions; the most common, G719S (COSM6252), is listed.

Mutation ^a	Exon	Chr	Position (hg19)	Ref	Alt	HGVS notation ^b
G719S (G719X)	18	7	55,241,707	C	T	p.Gly719Ser
E746_A750del	19	7	55,242,465	GGAATTAAGAGAAGCA	G	p.Glu746_Ala750del
T790M	20	7	55,249,071	C	T	p.Thr790Met
L858R	21	7	55,259,515	T	G	p.Leu858Arg

^a**Clinical shorthand** uses single-letter amino acid codes (e.g. E = Glu, A = Ala, T = Thr, L = Leu), as used in COSMIC and the oncology literature. ^b**HGVS notation** uses three-letter amino acid codes (e.g. Glu, Ala, Thr, Leu), the internationally standardised format used by SnpEff and ClinVar; see Table 25 for the SnpEff-annotated calls.

COSMIC identifiers: E746_A750del = COSM6223; L858R = COSM6224. Note: both callers returned a PASS call at position 55,249,063 (G→A, 8 bp upstream of T790M), annotated by SnpEff as the synonymous SNP `p.Gln787Gln` (dbSNP rs1050171) — a common germline polymorphism, not a somatic driver (see Section 8.2).

2 Background: Amplicon Sequencing

2.1 Workflow

Amplicon sequencing is a targeted NGS strategy in which only predefined genomic loci are sequenced [4]. The workflow has four steps:

1. **PCR amplification** — primers flanking the regions of interest selectively amplify those loci, producing many copies (*amplicons*) of each target.
2. **Adaptor ligation** — short synthetic adaptors are ligated to the PCR products, enabling flow-cell binding and cluster generation.
3. **Sequencing by synthesis (SBS)** — fluorescently labelled, reversible terminator nucleotides² are incorporated one at a time; a camera records which base was added at each cluster position per cycle.
4. **Base calling** — fluorescence images are decoded into FASTQ sequences with per-base Phred quality scores.

Because all sequencing capacity is focused on a tiny fraction of the genome, depths of 1,000–100,000× are routine, enabling detection of somatic mutations at <1% VAF that would be invisible at typical WGS depths (~30–100×). The trade-off is that only the designed targets are assayed — no off-target information is captured, and prior knowledge of the variants of interest is required.

²Each of the four bases (A, T, C, G) carries a distinct coloured dye so the camera can identify which base was just added. The “reversible terminator” chemical group blocks the strand from growing further until it is removed, ensuring exactly one base is incorporated per imaging cycle.

In this dataset, four amplicons cover EGFR hotspot exons 18–21, a design typical of **oncology panels** used to detect driver mutations in lung cancer.

2.2 Read Structure

Each read from an amplicon library consists of three elements (Figure 2):

Adaptor Synthetic oligonucleotide ligated to both fragment ends; enables flow-cell binding. Adaptors are not part of the biological sample and must be *trimmed* before analysis. This library uses two adaptors: AAGACTCGGCAGCATCTCCA and GCGATCGTCACTGTTCTCCA.

Primer Defines the amplicon boundaries; appears at the 5' end of R1 (forward) and R2 (reverse). Primers are trimmed or soft-clipped³ before variant calling to avoid boundary artefacts. Four primer pairs target one amplicon each (Table 4).

Amplified region The biological sequence of interest between the primers, carrying any variants subject to downstream analysis.

The expected read layout is:

[Adaptor] → [Fwd Primer] → [Amplified Region] → [Rev
Primer_{RC}] → [Adaptor_{RC}]

For short amplicons on a 2×250 bp MiSeq run, paired reads may overlap or read into the opposite adaptor, making adapter trimming especially important [5].

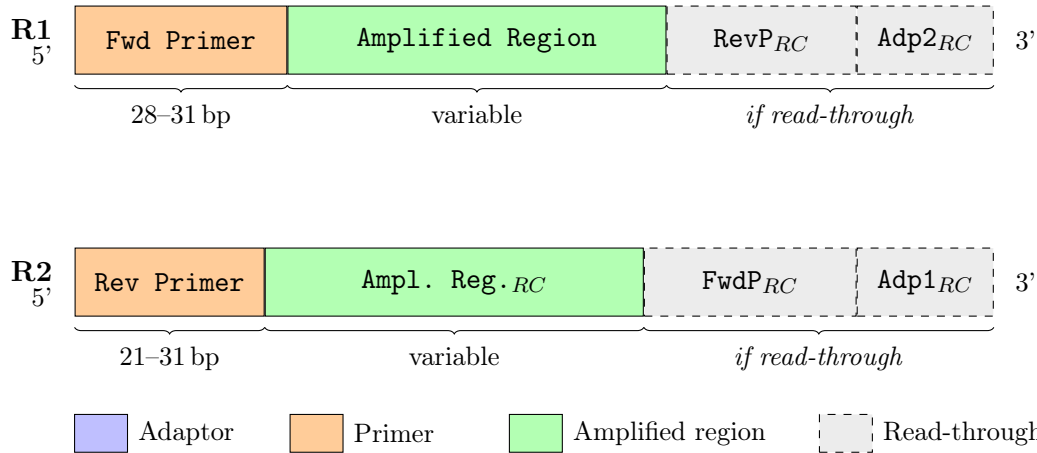


Figure 2: Expected structure of paired-end reads from MiSeq amplicon sequencing [5, 4]. The sequencing primer binds⁴ within the adaptor (Adp1 for R1, Adp2 for R2), so the adaptor itself is *not* part of the sequenced read; sequencing begins at the forward (R1) or reverse (R2) primer. Solid borders: segments always present. Dashed borders: segments that appear only when the read is long enough to sequence through the full insert (read-through). _{RC}: reverse complement.

³*Soft-clipping* (CIGAR code S) is a BAM alignment operation in which the aligner retains the full read sequence in the file but marks the primer bases as not contributing to the alignment score or the pileup, so they do not influence variant calls at those positions.

3 Dataset Description

This section documents the input files and library design parameters that were provided with the task, and characterises the upstream processing history extracted from the BAM file header.

3.1 Provided Files

Three files were provided:

- `aln.bam` — a pre-aligned BAM file mapped to the hg19 reference genome using NCBI-style chromosome names (e.g. 7 instead of `chr7`).
- `R1.fastq.gz` — raw paired-end reads, Read 1 (forward strand).
- `R2.fastq.gz` — raw paired-end reads, Read 2 (reverse strand).

3.2 Primer and Adaptor Sequences

The following primer pairs and adaptor sequences were provided with the task and define the library structure of each read (see Figure 2).

Amplicon	Forward Primer (5'→3')	Reverse Primer (5'→3')
1	TTGCCAGTTAACGTCTTCCTTCTCTCTCTG	GAGAAAAGGTGGGCCTGAGGTTTCAGAGCCA
2	CCCTTGTCTCTGTGTTCTTGTCCTCCCCCA	CCCCACCAGACCATGAGAGGCCCTGCGGCC
3	TGATCTGTCCCTCACAGCAGGGTCTTCTCT	TGACCTAAAGCCACCTCCTTA
4	CACACTGACGTGCCTCTCCCTCCCTCCA	CCGTATCTCCCTTCCCTGATTA

Table 4: Primer pairs defining the four EGFR amplicons. Reverse primers are given in 5'→3' orientation (i.e. their reverse complement appears at the 3' end of R1 reads for short inserts — read-through events).

Adaptor	Sequence (5'→3')	Length
Adaptor 1	AAGACTCGGCAGCATCTCCA	20 bp
Adaptor 2	GCGATCGTCACTGTTCTCCA	20 bp

Table 5: Custom adaptor sequences used in this dataset. Adaptor 1 is ligated to the 5' end of the insert; the Read 1 sequencing primer anneals within Adaptor 1 so the sequenced read begins with the forward primer, not the adaptor itself. Adaptor 2 is ligated to the opposite end and serves as the priming site for R2. The reverse complements of these adaptors appear in reads only upon read-through (Table 7).

3.3 BAM Header Analysis

A BAM file⁵ contains a plain-text header preceding the read data. The header is organised into typed records, each beginning with a two-letter tag:

⁵BAM (Binary Alignment Map) is the compressed binary form of the SAM (Sequence Alignment Map) format, the standard format for storing aligned sequencing reads.

@HD File-level metadata: SAM format version and sort order (`SO:unsorted` here, meaning reads are not sorted by coordinate).

@SQ Reference sequence dictionary: one entry per chromosome, giving the name and length used during alignment. This confirms the reference genome (hg19) and naming convention.

@RG Read group: sample identifier, sequencing platform, and library information. Here: `ID:Bordet_EGFR_S1, SM:S1, PL:Illumina`.

@PG Program record: the processing history of the file — which programs were run, in which order, with which version, and (where available) the exact command line used.

The @PG and @RG lines of this BAM are:

```
@RG ID:Bordet_EGFR_S1 SM:S1 PL:Illumina
@PG ID:bowtie2 PN:bowtie2 VN:2.0.2
```

This reveals three important facts:

- The reads were aligned with **bowtie2** (v2.0.2).
- **No adaptor trimming step is recorded** in the processing history. The raw FASTQs were passed directly to the aligner without prior Cutadapt or equivalent preprocessing.
- **No primer trimming step is recorded.** In a standard amplicon sequencing pipeline, `ivar trim` (or an equivalent tool) is run after alignment to soft-clip the PCR primer sequences from the 5' end of each read using a BED file of primer coordinates. This prevents the synthetic primer bases from entering the variant-calling pileup: because primer sequences are identical across every molecule from a given amplicon, any mismatch between the primer and the reference genome will appear as a near-100% VAF “variant” at the amplicon boundary — a primer-end artefact rather than a true somatic mutation. The absence of an `ivar trim` @PG entry confirms this step was skipped, with direct consequences for variant calling at amplicon boundaries (Section 8.2).

The consequence for adapter contamination is that reads containing adaptor sequence at their 3' end were aligned as-is; bowtie2 would have soft-clipped those bases, which may slightly reduce effective coverage at amplicon boundaries. Our Cutadapt run (Section 4) is therefore a purely retrospective QC characterisation and did not feed into the alignment.

The BAM header can be reproduced with the Snakemake rule `bam_header`:

```
bash pipeline/run.sh --cores 4 --until bam_header
# Output: pipeline/results/qc/bam_header.txt
```

4 Read Quality and Constitution

Question: What is the constitution of each read, for example (adaptor + primer + amplified region)?

Before characterising the specific structure of each read, we first verify that the raw data is of sufficient quality to analyse at all. General quality metrics (base quality, GC content, duplication) are independent of the library design and serve as a prerequisite sanity check — if the sequencing run itself failed, the downstream read constitution analysis would be meaningless. Only after confirming data quality do we proceed to examine the specific read layout using the known primer and adaptor sequences.

4.1 Raw Read Quality Assessment

4.1.1 Methods

FastQC [1] was run on the raw FASTQ files to assess overall read quality — base quality scores, GC content distribution, sequence length distribution, duplication levels, and adaptor content flags — before any library-specific analysis. FastQC was subsequently re-run on the adapter-trimmed reads as a before/after check.

```
bash pipeline/run.sh --cores 4 --until task1_fastqc_table
```

4.1.2 Results

Figure 3 shows the role of each tool in a standard NGS preprocessing pipeline.

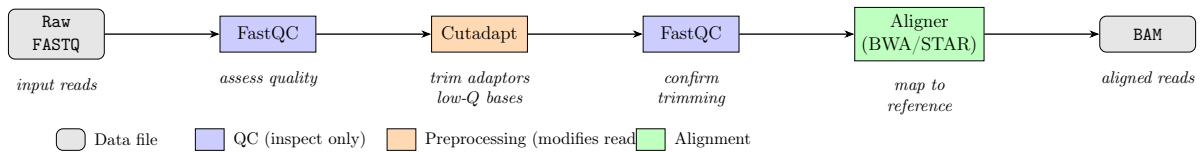


Figure 3: Typical NGS preprocessing pipeline before alignment. **FastQC** (blue) is a quality control tool — it inspects and reports read quality but does not modify any data. **Cutadapt** (orange) is a *preprocessing* tool — its primary purpose is to remove adaptor sequences and trim low-quality bases *before alignment*, so that the aligner receives clean reads and produces accurate mappings. A second FastQC run after trimming confirms that the cleaning was successful. In this dataset the BAM file was provided pre-aligned, so Cutadapt was run retrospectively on the raw FASTQs to confirm adaptor presence and measure insert size, rather than as a mandatory preprocessing step before mapping.

FastQC: Read Quality and Contamination Assessment

FastQC was run on both raw⁶ and adapter-trimmed⁷ reads; results are summarised in Table 6. All outputs are available in `pipeline/results/qc/`.

⁶*Raw reads* are the reads exactly as produced by the sequencer, before any processing. They contain the full sequence including adaptors, primers, low-quality bases, and the amplified insert.

⁷*Adapter-trimmed reads* are reads after Cutadapt has removed adaptor sequences from the 3' end and clipped low-quality bases ($Q < 20$). The biological insert (primer + amplified region) is left intact. Running FastQC on both raw and trimmed reads serves as a before/after check: adaptor warnings present before trimming should disappear afterwards, confirming that the correct sequences were specified and trimming worked as intended.

Module	Raw R1	Raw R2	Trim R1	Trim R2
Basic Statistics	pass	pass	pass	pass
Per base sequence quality	pass	warn	pass	pass
Per tile sequence quality	pass	pass	pass	pass
Per sequence quality scores	pass	pass	pass	pass
Per base sequence content	fail*	fail*	fail*	fail*
Per sequence GC content	fail*	fail*	fail*	fail*
Per base N content	pass	pass	pass	pass
Sequence Length Distribution	pass	pass	warn	warn
Sequence Duplication Levels	fail*	fail*	fail*	fail*
Overrepresented sequences	fail*	fail*	fail*	fail*
Adapter Content	pass	pass	pass	pass

* Expected artefact in amplicon sequencing — not a quality concern (see text).

Table 6: FastQC quality-control summary for raw and adapter-trimmed reads. Fail flags marked * are expected artefacts of targeted amplicon sequencing and do not indicate a quality problem.

The four **fail** flags marked with * are expected artefacts of amplicon sequencing and do not indicate a quality problem:

- **Per base sequence content:** FastQC expects a roughly equal A/T/G/C mix at each position. R1 always starts with the same forward primer (~30 bp of fixed sequence), so the first bases show a strongly biased composition that FastQC flags as a fail. This is normal for any targeted amplicon library. To confirm this is purely a primer artefact, one could hard-clip⁸ the primer bases before running FastQC (e.g. using `iVar trim` with the amplicon BED file): if the fail disappears, the bias originated solely from the fixed primer sequence; if it persists, it would indicate genuine amplification bias in the insert sequence itself.
- **Per sequence GC content:** A whole-genome library produces a smooth bell-curve GC distribution centred on the genome average. Here only four specific amplicons are sequenced, each with its own fixed GC content, producing a multi-modal or shifted distribution. The per-primer GC content is analysed in detail in Section 7.2.
- **Sequence duplication levels:** PCR amplification generates many identical copies of the same molecule by design. High duplication is expected and intentional (deep coverage is the goal); it would only be a concern in WGS.
- **Overrepresented sequences:** Primer and amplicon sequences repeat thousands of times across reads by construction.

The modules that *do* matter — per-base quality, per-tile quality, per-sequence quality scores, per-base N content, and adapter content — are mostly clean, with two non-critical observations:

⁸*Hard-clipping* (CIGAR code H) permanently removes bases from both the read sequence and the BAM record — unlike soft-clipping (S), which retains the bases in the file but excludes them from the alignment. Hard-clipped bases are gone and cannot be recovered from the BAM.

- **Per-base sequence quality, raw R2 (warn):** Phred scores decline at the 3' end of R2, a common sequencing-by-synthesis artefact (later cycles accumulate more signal noise). Crucially, the BAM file provided was aligned from the *raw* reads without a prior quality-trimming step (Section 3.3), so these lower-quality bases are present in the alignment. All downstream variant callers apply a per-base quality threshold of $Q \geq 20$ (VarDictJava `-q 20`, VarScan2 `-min-avg-qual 20`, samtools depth `-Q 20`), which excludes the worst bases. The warn resolves to pass after Cutadapt trimming, confirming that quality trimming before alignment would have fully mitigated this issue.
- **Sequence length distribution, trimmed reads (warn):** Adapter trimming produces reads of variable length (some reads are shorter after trimming), which FastQC flags whenever lengths are not uniform. This is expected behaviour and does not affect downstream analysis.

4.2 Read Constitution

4.2.1 Methods

Having confirmed the data quality, we characterise the specific structure of each read using the primer and adaptor sequences documented in Section 3 (Tables 4 and 5). This addresses four practical questions:

1. **Adaptor identification** — which adaptor sequences flank the amplicons, and what is the correct trimming strategy?
2. **Adaptor contamination** — what fraction of reads contain read-through adaptor sequence at the 3' end?
3. **Primer presence** — do the primer sequences appear at the expected 5' position in each read, confirming the amplification worked as designed?
4. **Fragment length** — what are the insert sizes per amplicon, and are they consistent with the amplicons being longer than the read length?

The following steps were carried out:

1. A custom script searched for all four adaptor sequences (Adp1, Adp1_{RC}, Adp2, Adp2_{RC}) in the raw reads to identify which adaptors flank the amplicons and confirm the correct trimming sequences (Section 4.2.2).
2. **Cutadapt** [11] was run with the confirmed adaptor sequences to quantify adaptor contamination rates in the reads as a QC step (Section 4.2.2).
3. Primer sequences were verified to be present at the expected 5' position of each read by primer-matching assignment (Section 4.2.3; per-amplicon read-through detail in Appendix D).
4. Per-amplicon insert size distributions were extracted from the BAM TLEN field to confirm each amplicon's fragment length (Section 4.2.4).

4.2.2 Adaptor verification: which adaptor appears in each read?

Where do adaptors appear in a read? Although every sequenced DNA molecule carries adaptors on *both* ends, only the **3' end** can contain adaptor sequence in the read itself (see Figure 2 for the full read layout). The 5' adaptor is the sequencing primer binding site — sequencing starts from it, so it is never incorporated into the read sequence. The 3' adaptor (on the far end of the molecule) only appears in a read when the insert is shorter than the read length: the sequencer reads through the entire amplicon and continues into the far adaptor. This *read-through* is therefore the sole source of adaptor contamination, and it can only ever appear at the **3' end** of the read.

Two library configurations are possible depending on whether the same adaptor or two distinct adaptors flank the amplicon (Figure 4). Since this dataset provides **two distinct** adaptor sequences ($\text{Adp1} \neq \text{Adp2}$), **Scenario B** (different adaptors on each end) was assumed; this was subsequently confirmed empirically (Appendix C).

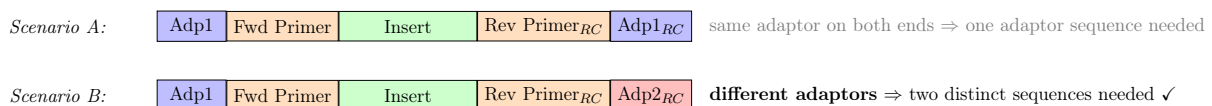


Figure 4: The two possible adaptor configurations for a paired-end amplicon library. Scenario A uses the same adaptor on both ends (one sequence suffices); Scenario B uses two distinct adaptors (two sequences required). Since two distinct adaptor sequences ($\text{Adp1} \neq \text{Adp2}$) were provided with this dataset, Scenario B was assumed and subsequently confirmed empirically (Appendix C).

Is there adapter contamination? Adapter contamination refers to adapter sequences appearing within reads, which occurs when the sequencer reads through the entire insert and into the far adaptor. A small fraction of contaminated reads is expected in any paired-end library and is not a quality failure — it simply indicates that those fragments are shorter than the read length. Using the Snakemake rule `task1_check_adaptors` (`bash pipeline/run.sh -until task1_check_adaptors`), the script `check_adaptors.py` searched for all four sequences (Adp1 , Adp1_{RC} , Adp2 , Adp2_{RC}) in the raw reads (Table 7). The results match expectations precisely:

- **5' priming adaptors absent from reads.** Adp1 in R1⁹ (0.682%) and Adp2 in R2 ($\approx 0\%$) are effectively absent, as expected — they are the primer binding sites from which sequencing starts and are therefore never incorporated into the read sequence.
- **RC sequences present at the 3' end.** Adp2_{RC} appears in 4.8% of R1 reads and Adp1_{RC} in 5.3% of R2 reads — the read-through signal from the far adaptor. This rate is consistent with the insert size distributions showing all four amplicons peak well above 150 bp (Figure 5). The asymmetry between R1 (4.8%) and R2 (10.9%) suggests that the reverse primer sits closer to the far adaptor than the forward primer — a common consequence of primers being placed at different offsets relative to the region of interest.

The presence of *different* RC signals in R1 and R2 (Adp2_{RC} vs. Adp1_{RC}) confirms **Scenario B** (different adaptors on each end), which determines the correct Cutadapt trim-

⁹R1 (Read 1) and R2 (Read 2) are the two reads of a paired-end run, as defined in Section 3.

ming flags used in Section 4.2.2. The correct trimming strategy is therefore to clip Adp2_{RC} from the 3' end of R1 and Adp1_{RC} from the 3' end of R2, as configured with Cutadapt's `-a/-A` flags.

The full derivation of how the two scenarios produce distinguishable read-through fingerprints, and how the script exploits this to confirm Scenario B, is given in Appendix C.

Label	Sequence (5'→3')	R1 count	R1 %	R2 count	R2 %
Adp1 (fwd)	AAGACTCGGCAGCATCTCCA	3,924	0.682%	0	0.000%
Adp2 (fwd)	GCGATCGTCACTGTTCTCCA	7	0.001%	2	0.000%

Table 7: Occurrence of both adaptor sequences and their reverse complements (RC) in R1 and R2 raw reads. Bold values indicate the dominant signal in each read. Adp2 RC is dominant in R1 and Adp1 RC is dominant in R2, confirming Scenario B: the library uses different adaptors on each end (standard paired-end).

Cutadapt trimming Cutadapt is often used to remove adaptor contamination: it searches for the confirmed adaptor sequences at the 3' end of each read and clips them, along with any low-quality bases. The trimming rates (Table 8) directly quantify how many reads were affected by read-through contamination.

Metric	Count	Fraction
Total read pairs processed	575,002	
Read 1 with Adaptor 1 detected	35,489	(6.2%)
Read 2 with Adaptor 2 detected	62,613	(10.9%)
Pairs discarded (too short)	1,983	(0.3%)
Pairs written (passing filters)	573,019	(99.7%)
Total bases quality-trimmed	2,122,235	(1.2%)

Table 8: Cutadapt [11] adapter trimming summary for the paired-end MiSeq dataset. Adaptor 1 (TGGAGAACAGTGACGATCGC) trimmed from R1; Adaptor 2 (TGGAGATGCTGCCGAGTCTT) trimmed from R2.

Adapter detection rates of 6.2% on R1 and 10.9% on R2 indicate that the majority of reads do *not* read through into the opposite adapter. This confirms that most amplicons are **longer than the read length** (~ 150 bp), so the majority of reads terminate within the amplified region before reaching the far adapter. The 6–11% of adapter-containing reads likely originate from shorter amplicons or degraded fragments where the read length exceeds the insert size. The higher rate on R2 is consistent with reads from the reverse direction reaching the forward adapter more frequently.

Quality trimming removed a small fraction of bases from the 3' ends, suggesting overall good base quality across the run. 99.7% of read pairs passed all filters and were written to the trimmed output.

Conclusion. Adaptor trimming affected only a small fraction of reads: roughly 6% of R1 and 11% of R2 reads had adaptor sequence removed; the remaining ~ 90 –94% were unchanged in sequence content.

Importantly, Cutadapt was *not* part of the upstream pipeline that generated the provided BAM file — the BAM header confirms that bowtie2 was run directly on the raw reads without a prior trimming step (Section 3.3). This likely had little practical impact on the alignment results: bowtie2 performs *soft-clipping* by default, meaning adaptor bases at the 3' end of a read are excluded from the alignment score rather than causing a mapping failure. Given that only ~6–11% of reads carried any adaptor sequence, and those bases would have been soft-clipped by the aligner, the effect on coverage and variant calls is expected to be minimal. This contamination rate was also confirmed at the per-amplicon level, with all four amplicons showing read-through rates below 5% (Appendix D).

4.2.3 Primer verification

Having confirmed adaptor presence, the next QC question is whether the primer sequences themselves are present at the expected 5' position in each read. In amplicon sequencing every read should begin with one of the four known primer sequences; absence of a primer in a read indicates that the fragment may be degraded, misassigned, or a primer-dimer artefact.¹⁰

Each read pair was assigned to an amplicon by matching the first 20 bp of R1 against the forward primer sequences (up to 2 mismatches); if no match was found in R1, R2 was matched against the reverse primer sequences. To reproduce:

```
bash pipeline/run.sh --cores 4 --until
    task1_primer_assignment_table
```

Amplicon	Forward primer (first 20 bp)	Read pairs	%
Amplicon 1	TTGCCAGTTAACGTCTTCCT	227,897	39.6%
Amplicon 2	CCCTTGTCTCTGTGTTCTTG	10,948	1.9%
Amplicon 3	TGATCTGTCCCTCACAGCAG	110,440	19.2%
Amplicon 4	CACACTGACGTGCCTCTCCC	202,062	35.1%
Unassigned	—	23,655	4.1%
Total		575,002	100.0%

Table 9: Primer assignment rates. Each read pair was assigned to an amplicon by matching the first 20 bp of R1 against the known forward primer sequences (up to 2 mismatches); if no match was found in R1, R2 was matched against the reverse primer sequences. The assignment rate per amplicon confirms that primer sequences are present at the expected 5' position of the reads. Unassigned pairs are reads whose 5' end did not match any known primer.

¹⁰A *primer-dimer* is a short artefactual PCR product that forms when the 3' end of one primer has even partial complementarity (as few as 4–6 bp) to another primer (or to itself) and is extended by DNA polymerase instead of annealing to the template. Full complementarity is not required: because G–C pairs contribute three hydrogen bonds (versus two for A–T), GC-rich 3' ends form stable partial duplexes at relatively low complementarity. The resulting fragment contains only primer sequence with no intervening genomic content. In a multiplex reaction, the many primers present at high concentration increase the chance of such interactions. In sequencing, a primer-dimer read begins with the expected forward primer (and is therefore assigned to an amplicon) but its remaining sequence is the reverse complement of another primer rather than the amplified locus, so it cannot be aligned to the reference genome.

95.9% of read pairs were successfully assigned to a primer, confirming that primer sequences are present at the expected 5' position in the vast majority of reads. The large imbalance between Amplicon 2 (1.9%, 10,948 pairs) and the other three amplicons (~35–40%) is notable and is discussed further in the coverage analysis (Section 7).

Unassigned reads (4.1%). Reads are unassigned when the first 20 bp of neither R1 nor R2 matches any known primer within 2 mismatches. Possible causes include:

- **Primer-dimer artefacts** — during PCR, primers can ligate to each other rather than anneal to the template, producing short fragments whose 5' end is a partial or mismatched primer sequence.
- **Degraded DNA** — if the original template was fragmented, some reads may start mid-amplicon (after the primer region) and therefore have no primer at position 1.
- **Non-specific amplification** — off-target products that do not begin with any of the four known primers.
- **Sequencing errors at the 5' end** — the first few bases of Illumina reads tend to have slightly lower quality; sufficient errors in the first 20 bp can push an otherwise matching read over the 2-mismatch threshold.

Without further lab tests it is not possible to determine which cause dominates; the 4.1% rate is noted as a QC observation. Notably, unassigned reads are disproportionately represented in the unmapped fraction: 55% of all unassigned read pairs fail to align, compared to 8.3% across all reads (Table 20), consistent with these reads not originating from the target amplicons at all.

4.2.4 Per-amplicon insert size distribution

The *insert size* is the length of the original DNA fragment between the two sequencing adaptors — the amplified region itself, excluding the adaptors and primers (see Figure 2 for the full read layout). In a paired-end BAM file the insert size is recorded as the TLEN field (template length) of each read.

The TLEN of each properly paired read pair was extracted from the pre-aligned BAM using `samtools view` and grouped by amplicon using the BLAST-derived coordinates in `amplicons.bed`. Figure 5 shows the resulting per-amplicon histograms. The red dotted vertical line marks the 150 bp read length; amplicons whose distribution peak lies to the left of this line would produce read-through into the adaptor sequence on both R1 and R2.

To reproduce:

```
bash pipeline/run.sh --cores 4 --until insert_size_plot
```

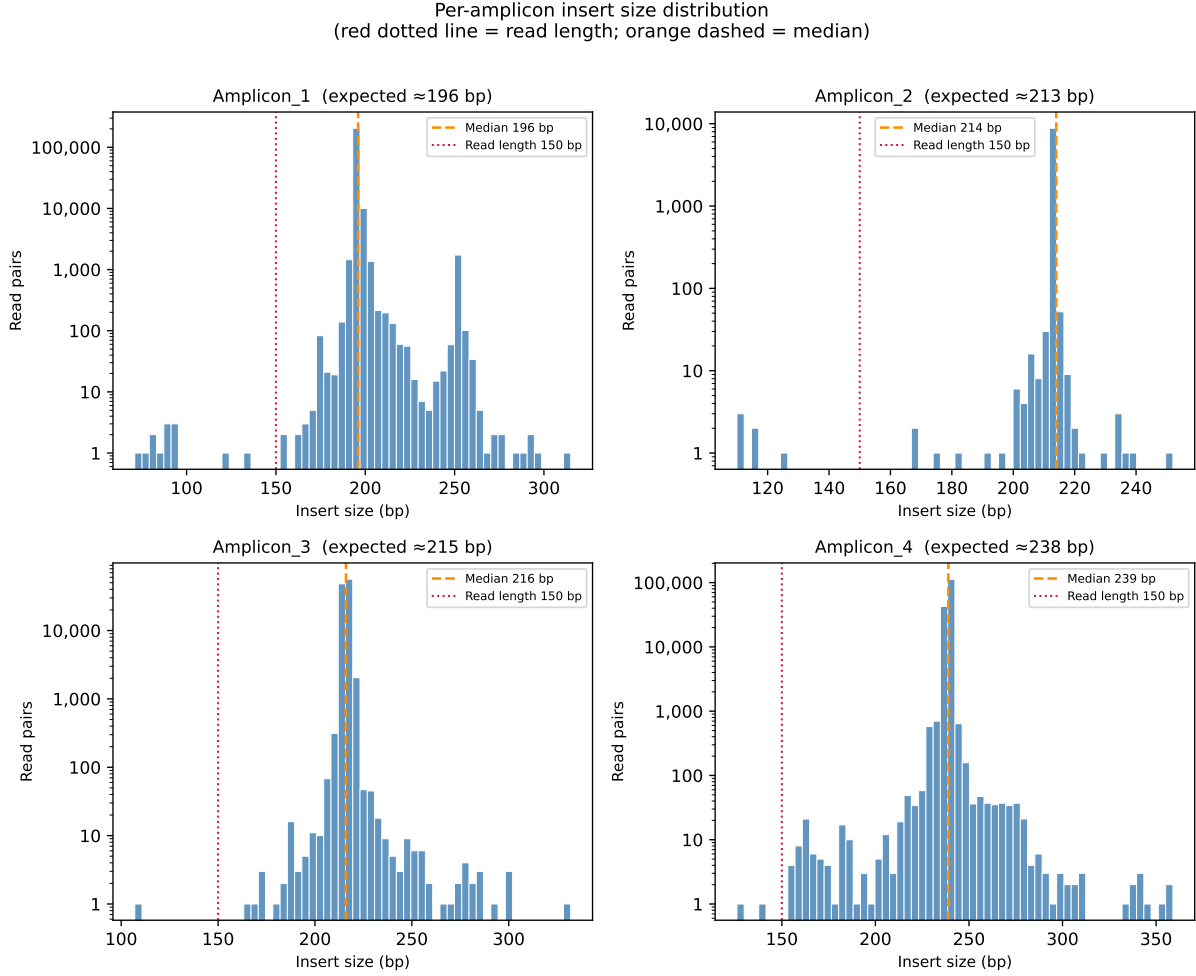


Figure 5: Per-amplicon insert size distributions derived from the TLEN field of properly paired, mapped reads (`samtools view -f 2 -F 2308`). The y-axis is on a log scale to reveal low-frequency secondary peaks (e.g. primer-dimers, off-target products) that would be invisible on a linear scale given the dominant sharp peak of on-target amplicon fragments. Orange dashed line: median insert size for that amplicon. Red dotted line: read length (150 bp) — amplicons with a distribution peak below this line generate read-through into the adaptor sequence. All four amplicons peak well above 150 bp, consistent with the low read-through rates in Table 27.

4.2.5 Summary: read constitution

Combining the findings above, each read has the following structure:

[Primer (~ 21 -30 bp)] $\rightarrow$ [Amplified region (135-187 bp insert)] $\rightarrow$ [Adp_{RC} (3' end, ~ 6 -11% of reads only)]

Primer sequences were confirmed at the expected 5' position in 95.9% of read pairs (Section 4.2.3). The amplified region spans 135–187 bp of insert sequence (Table 13), giving total amplicon lengths of 195–237 bp — all longer than the 150 bp read length, so R1 and R2 overlap in the centre of the amplicon rather than reading through into the opposite adaptor. Adaptor sequence at the 3' end was observed in $\sim 6\%$ of R1 reads and $\sim 11\%$ of R2 reads (Section 4.2.2), originating from a minority of shorter fragments (degraded DNA, primer-dimers) where the read length exceeds the insert size. This confirms the

expected amplicon library structure illustrated in Figure 2.

5 Gene Region Mapping

Question: Which gene region did those reads map to?

Two complementary approaches were used to identify the targeted gene regions: first, the primer sequences were aligned directly to hg19 using BLAST to establish the designed amplicon coordinates; second, read coverage was computed from the BAM to confirm those intervals with actual data and annotate overlapping genes.

5.1 Primer Localisation by BLAST

Methods. The eight primer sequences (four forward, four reverse; from `pipeline/config.yaml`) are written to a FASTA file by the Snakemake rule `blast_primers`:

```
>amplicon1_fwd    TTGCCAGTTAACGTCTTCCTTCTCTCTCTG
>amplicon1_rev    GAGAAAAGGTGGGCCTGAGGTTTCAGAGCCA
>amplicon2_fwd    CCCTTGTCTCTGTGTTCTTGTCCCCCCCCA
>amplicon2_rev    CCCACCCAGACCATGAGAGGCCCTGCGGCC
>amplicon3_fwd    TGATCTGTCCCTCACAGCAGGGTCTTCTCT
>amplicon3_rev    TGACCTAAAGCCACCTCCTTA
>amplicon4_fwd    CACACTGACGTGCCTCTCCCTCCCTCCA
>amplicon4_rev    CCGTATCTCCCTTCCCTGATTA
```

These are queried against a local BLAST nucleotide database built from the hg19 reference FASTA (`hg19.fa.gz`, UCSC Genome Browser) using `makeblastdb`. The search uses parameters tuned for short sequences:

```
blastn -query primers.fa -db hg19 \
  -task blastn-short \
  -word_size 7 \
  -evaluate 1e-3 \
  -perc_identity 90 \
  -max_target_seqs 1 \
  -outfmt "6 qseqid sseqid pident length mismatch gapopen \
          qstart qend sstart send evaluate bitscore sstrand"
```

Key parameter choices:

- `-task blastn-short`: preset that lowers the word size and adjusts scoring for sequences <50 bp. Standard `blastn` uses word size 11, which is too large to seed alignments reliably for 20–30 bp primers.
- `-word_size 7`: shorter seed increases sensitivity; acceptable here because we only query 8 sequences.
- `-evaluate 1e-3`: permissive given the short query length; a 20 bp perfect match against a 3 Gbp genome has an E-value well below this cutoff.
- `-perc_identity 90`: requires $\geq 90\%$ identity, filtering spurious low-identity hits while tolerating one or two mismatches.

- `-max_target_seqs 1`: reports only the top genomic hit per primer, giving the primary binding site.

Each amplicon’s genomic span is defined as the interval from the forward primer hit start to the reverse primer hit end on the same chromosome. The two one-time setup rules and the search rule are:

```
# One-time setup (download ~1 GB, build BLAST index):
bash pipeline/run.sh --cores 4 --until build_blast_db

# Run primer alignment:
bash pipeline/run.sh --cores 4 --until blast_primers
```

Amplicon	Chr	Start (hg19)	End (hg19)	Length	Fwd id.	Rev id.
Amplicon 1	chr7	55,242,379	55,242,574	195 bp	100%	100%
Amplicon 2	chr7	55,241,584	55,241,796	212 bp	100%	100%
Amplicon 3	chr7	55,259,375	—	—	100%	no hit
Amplicon 4	chr7	55,248,957	55,249,194	237 bp	100%	100%

Table 10: Primer localisation by `blastn` against hg19 (`-task blastn-short`, `-perc_identity 90`, `-evalue 1e-3`). Start and End define the amplicon span from forward primer start to reverse primer end. “Fwd id.” and “Rev id.” are the percent identity of the best BLAST hit; “no hit” indicates the primer fell below the E-value threshold.

Results. All four forward primers align to **chr7** with 100% identity, confirming that the panel targets **EGFR** (Epidermal Growth Factor Receptor) on chromosome 7. The primer coordinates span EGFR exons 18–21 (hg19 positions ~55.24–55.26 Mb), the kinase domain hotspot region most commonly mutated in non-small-cell lung cancer.

The reverse primer for Amplicon 3 produced no hit at the $E < 10^{-3}$ threshold; this primer is only 21 bp (the shortest in the panel) and its best genomic match has $E = 0.003$, just above the cutoff. As shown in the off-target search (Table 12), this hit is on chr7 with 100% identity over all 21 bp — it is the correct EGFR binding site, merely filtered by the strict threshold. Table 11 combines both searches to give complete coordinates for all four amplicons.

Amplicon	Chr	Start (hg19)	End (hg19)	Length	Fwd id.	Rev id.
Amplicon 1	chr7	55,242,379	55,242,574	195 bp	100%	100%
Amplicon 2	chr7	55,241,584	55,241,796	212 bp	100%	100%
Amplicon 3	chr7	55,259,375	55,259,589	214 bp	100%	100% [†]
Amplicon 4	chr7	55,248,957	55,249,194	237 bp	100%	100%

Table 11: Complete primer localisation combining the strict search (`-evalue 1e-3`) and the permissive off-target search. All four amplicons are resolved to chr7 (EGFR), including Amplicon 3 whose reverse primer was below the strict E-value threshold. [†] Hit recovered from the permissive off-target search (`-evalue 1`; Table 12); E-value just exceeded the strict 10^{-3} cutoff.

Off-target search (permissive BLAST). To check whether any primer could produce off-target amplification elsewhere in the genome, the search was repeated with permissive parameters (rule `blast_primers_offtarget`): `-max_target_seqs 10`, `-perc_identity 70`, `-evalue 1`. Table 12 lists all secondary hits (primary EGFR hit per primer excluded). All secondary hits have alignment lengths of 17–20 bp and E-values of 0.003–0.80 — consistent with background noise rather than specific secondary binding sites. Crucially, **none of these partial hits fall near chr4:77.6 Mb**, ruling out off-target primer priming at the SHROOM3 locus observed in the coverage analysis below.

Primer	Chr	Aln len (bp)	%id	E-value	Strand	Type
amplicon1_fwd	chr7	30	100	3.35e-08	plus	<i>on-target</i>
	chr19	18	100	0.49	plus	<i>off-target</i>
	chrX	18	100	0.49	minus	<i>off-target</i>
amplicon2_fwd	chr7	29	100	1.21e-07	plus	<i>on-target</i>
	chr7	18	100	0.44	plus	<i>off-target</i>
	chrY	18	100	0.44	plus	<i>off-target</i>
	chrY	18	100	0.44	minus	<i>off-target</i>
amplicon3_fwd	chr7	30	100	3.35e-08	plus	<i>on-target</i>
	chr7	20	100	0.03	minus	<i>off-target</i>
	chr11	20	100	0.03	minus	<i>off-target</i>
	chr3	20	100	0.03	plus	<i>off-target</i>
	chr7	19	100	0.12	minus	<i>off-target</i>
	chr4	19	100	0.12	minus	<i>off-target</i>
amplicon4_fwd	chr7	28	100	4.36e-07	plus	<i>on-target</i>
	chr14	19	100	0.10	minus	<i>off-target</i>
	chr14	18	100	0.40	plus	<i>off-target</i>
	chr14	18	100	0.40	plus	<i>off-target</i>
	chr22	18	100	0.40	plus	<i>off-target</i>
	chr17	22	95	0.40	plus	<i>off-target</i>
	chr17	18	100	0.40	minus	<i>off-target</i>
	chr10	18	100	0.40	plus	<i>off-target</i>
	chrX	18	100	0.40	plus	<i>off-target</i>
	chrX	18	100	0.40	minus	<i>off-target</i>
	chr6	18	100	0.40	plus	<i>off-target</i>
amplicon1_rev	chr7	30	100	3.35e-08	minus	<i>on-target</i>
	chr15	20	100	0.03	minus	<i>off-target</i>
	chr11	22	95	0.49	plus	<i>off-target</i>
amplicon2_rev	chr7	30	100	3.35e-08	minus	<i>on-target</i>
	chr7	19	100	0.12	minus	<i>off-target</i>
	chr18	18	100	0.49	plus	<i>off-target</i>
	chr13	18	100	0.49	plus	<i>off-target</i>
amplicon3_rev	chr7	21	100	3.00e-03	minus	<i>on-target (relaxed threshold)</i>
	chr1	19	100	0.04	plus	<i>off-target</i>
	chr6	18	100	0.16	minus	<i>off-target</i>
	chr13	17	100	0.64	minus	<i>off-target</i>
	chrX	17	100	0.64	minus	<i>off-target</i>
amplicon4_rev	chr7	22	100	8.29e-04	minus	<i>on-target</i>
	chr10	17	100	0.80	plus	<i>off-target</i>
	chr6	17	100	0.80	minus	<i>off-target</i>
	chr4	17	100	0.80	minus	<i>off-target</i>

Table 12: All hits from the permissive off-target BLAST search (-perc_identity 70, -evalue 1, -max_target_seqs 10). **Bold** rows are the best hit per primer: ***on-target*** primers were also found in the strict search (Table 10); ***on-target (relaxed threshold)*** indicates the primer's best hit is the correct EGFR site but its E-value (0.003) just exceeded the strict cutoff of 10^{-3} . Normal-weight rows are secondary *off-target* hits sorted by E-value within each primer; all have alignment lengths of 17–21 bp and E-values ≥ 0.003 , consistent with background noise. No off-target hit falls near chr4:77.6 Mb (SHROOM3 locus).

Amplicon structure and read coverage. Each amplicon sequenced on the MiSeq consists of three contiguous parts: the **forward primer**, the **biological insert** (the region of interest carrying any variants), and the **reverse primer**. The total amplicon length reported by BLAST is therefore $l_{\text{amp}} = l_{\text{fwd}} + l_{\text{insert}} + l_{\text{rev}}$ (Figure 6a).

In paired-end sequencing, R1 reads inward from the forward primer and R2 reads inward from the reverse primer (Figure 6b). Because all four amplicons are *longer* than the read length ($l_{\text{amp}} = 195\text{--}237\text{ bp} > r = 150\text{ bp}$), each read terminates inside the amplicon before reaching the far adapter — explaining the near-zero read-through rates ($<5\%$) observed in Section 4. At the same time, because all amplicons are shorter than twice the read length ($l_{\text{amp}} < 2r = 300\text{ bp}$), R1 and R2 do meet in the centre, overlapping by $2r - l_{\text{amp}}$ bases (Table 13).

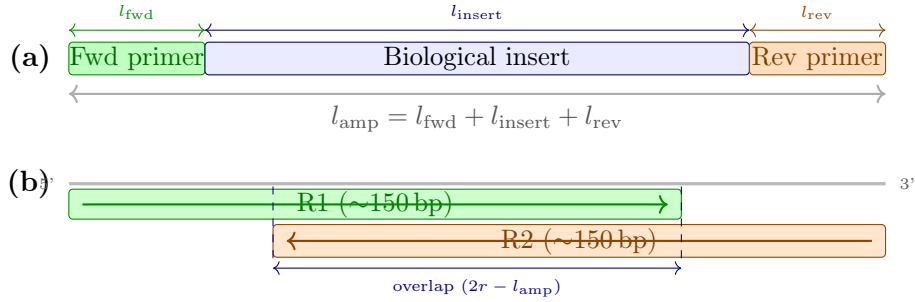


Figure 6: (a) An amplicon consists of the forward primer, the biological insert, and the reverse primer. The total amplicon length is $l_{\text{amp}} = l_{\text{fwd}} + l_{\text{insert}} + l_{\text{rev}}$; BLAST primer localisation reports l_{amp} directly as the span from forward primer start to reverse primer end. (b) R1 reads inward from the forward primer; R2 reads inward from the reverse primer. When $l_{\text{amp}} < 2r$ (where $r \approx 150\text{ bp}$ is the read length), the two reads overlap in the centre by $2r - l_{\text{amp}}$ bases. Schematic proportions; not to exact scale.

Amplicon	Amp. length	Fwd primer	Insert	Rev primer	R1/R2 overlap
Amplicon 1	195 bp	30 bp	135 bp	30 bp	105 bp
Amplicon 2	212 bp	29 bp	153 bp	30 bp	88 bp
Amplicon 3	214 bp	30 bp	163 bp	21 bp	86 bp
Amplicon 4	237 bp	28 bp	187 bp	22 bp	63 bp

Table 13: Amplicon composition derived from BLAST primer coordinates (Table 11) and primer sequences (Table 4). Insert length = $l_{\text{amp}} - l_{\text{fwd}} - l_{\text{rev}}$. R1/R2 overlap = $2r - l_{\text{amp}}$ where $r = 150\text{ bp}$; positive values indicate the two reads overlap in the centre of the amplicon (see Figure 6).

Gene annotation of targeted amplicons. To confirm which genes the four amplicons target, the BLAST-derived amplicon coordinates (Table 11) were intersected with the hg19 RefFlat gene annotation using `bedtools intersect -wo`. Table 14 shows that all four amplicons map entirely within the *EGFR* gene (100% overlap), confirming that the panel is a targeted EGFR sequencing assay. Amplicon 4 additionally overlaps *EGFR-AS1*, a long non-coding antisense RNA whose locus is embedded within the EGFR gene body on the opposite strand; this is an expected annotation artefact rather than an off-target design.

Background: EGFR and EGFR-AS1. *EGFR* (Epidermal Growth Factor Receptor; chr7p11.2) encodes a transmembrane receptor tyrosine kinase of the ErbB family. Upon ligand binding, EGFR homo- or heterodimerises and activates downstream proliferation and survival pathways (RAS/MAPK, PI3K/AKT) [16]. Somatic activating mutations in EGFR — most commonly in-frame deletions in exon 19 and the L858R point mutation in exon 21 — are driver events in non-small-cell lung cancer (NSCLC) and confer sensitivity to EGFR tyrosine kinase inhibitors such as gefitinib and erlotinib [10]. Each amplicon covers exactly one exon of the kinase-domain hotspot: Amplicon 2 covers exon 18 (G719 mutations), Amplicon 1 covers exon 19 (common in-frame deletions), Amplicon 4 covers exon 20 (insertion hotspot), and Amplicon 3 covers exon 21 (L858R substitution). Together the four amplicons span the full exon 18–21 mutational hotspot, consistent with a clinical EGFR variant-detection assay (Figure 7).

EGFR-AS1 (EGFR Antisense RNA 1) is a long non-coding RNA (lncRNA) transcribed from the antisense strand of the EGFR locus and annotated as a separate gene entry in RefSeq. Because it occupies the same genomic interval as EGFR on the opposite strand, any amplicon spanning that region will intersect both entries in a strand-agnostic BED intersection — hence the dual annotation for Amplicon 4. EGFR-AS1 has been reported to modulate EGFR mRNA levels in *cis*, but its functional role remains an active area of research; it is not a target of this sequencing panel.

Amplicon	Gene	Amp. len (bp)	Overlap (bp)	Overlap (%)
Amplicon 1	EGFR	196	196	100.0
Amplicon 2	EGFR	213	213	100.0
Amplicon 3	EGFR	215	215	100.0
Amplicon 4	EGFR	238	238	100.0
	EGFR-AS1	238	238	100.0

Table 14: Gene overlaps for the four BLAST-defined amplicon intervals (Table 11), computed with `bedtools intersect -wo` against the hg19 RefFlat gene annotation. Overlap is reported in base pairs and as a percentage of the amplicon length.

Table 15 lists the four targeted exons with their exact hg19 coordinates and the associated somatic mutations relevant to NSCLC diagnosis and treatment. Exons 19 and 21 together account for approximately 85% of all oncogenic EGFR alterations; exon 18 G719X and exon 20 insertions make up most of the remainder. The panel therefore covers the full spectrum of clinically actionable EGFR variants screened in routine oncology practice.

Exon	Amplicon	Start (hg19)	End (hg19)	Len (bp)	Known somatic mutations
18	Amplicon 2	55,241,613	55,241,736	123	G719X point mutations; uncommon ($\sim 3\%$ of EGFR mutations), sensitive to afatinib and osimertinib
19	Amplicon 1	55,242,414	55,242,513	99	In-frame deletions (e.g. E746_A750del); most common EGFR alteration ($\sim 45\%$), strongly sensitive to EGFR TKIs
20	Amplicon 4	55,248,985	55,249,171	186	In-frame insertions; generally confer resistance to first/second-generation TKIs; targeted by amivantamab and mobocertinib
21	Amplicon 3	55,259,411	55,259,567	156	L858R point mutation (c.2573T>G); second most common EGFR alteration ($\sim 40\%$), sensitive to EGFR TKIs

Table 15: EGFR kinase-domain exons targeted by the amplicon panel (NM_005228, hg19, plus strand). Coordinates are 0-based half-open (BED convention), derived from the UCSC hg19 refFlat table. Exons 18–21 together account for $\gtrsim 85\%$ of all oncogenic EGFR mutations and are the primary predictive biomarkers for tyrosine kinase inhibitor (TKI) therapy in non-small-cell lung cancer (NSCLC) [10].

Amplicon coverage of target exons. Each amplicon fully contains its target exon: the primers are placed in the flanking intronic sequence on either side, so the sequenced region spans a short intronic flank, the complete exon, and another short intronic flank. Table 16 gives the exact amplicon and exon coordinates alongside the left and right intronic flanks for each amplicon.

Exon	Amplicon	Amp. start	Amp. end	Amp. len (bp)	Exon start	Exon end	Exon len (bp)
18	Amplicon 2	55,241,583	55,241,796	213	55,241,613	55,241,736	123
19	Amplicon 1	55,242,378	55,242,574	196	55,242,414	55,242,513	99
20	Amplicon 4	55,248,956	55,249,194	238	55,248,985	55,249,171	186
21	Amplicon 3	55,259,374	55,259,589	215	55,259,411	55,259,567	156

Table 16: Amplicon spans relative to their target EGFR exon (NM_005228, hg19, plus strand, 0-based half-open coordinates). Each amplicon fully contains its target exon; the primers are placed in the flanking intronic sequence on either side. Left intronic flank = exon start – amplicon start; right intronic flank = amplicon end – exon end.

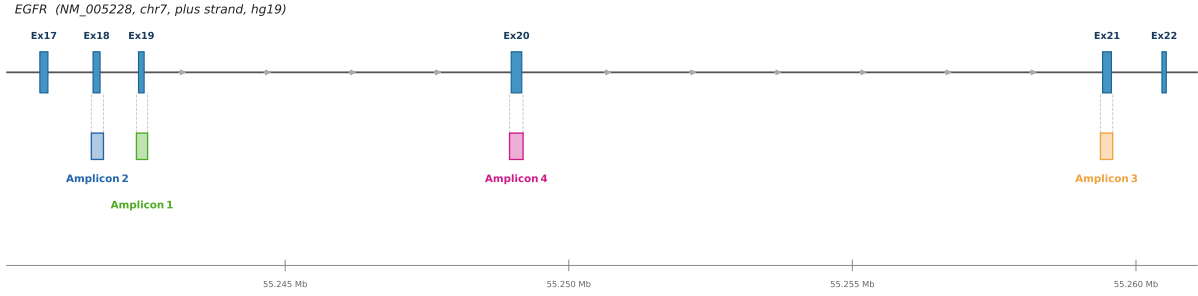


Figure 7: Exon structure of EGFR (NM_005228, hg19, chr7 plus strand) in the region targeted by the amplicon panel. Blue boxes indicate exons; exon numbers follow the canonical transcript order (5' to 3', left to right). The four amplicons (coloured bars) were localised by BLAST primer alignment (Table 11). Each amplicon spans exactly one exon: Amplicon 2 covers exon 18 (G719 mutation site), Amplicon 1 covers exon 19 (the most common site of in-frame deletions in NSCLC [10]), Amplicon 4 covers exon 20 (insertion hotspot), and Amplicon 3 covers exon 21 (L858R substitution site [10]). Together the four amplicons span the full kinase-domain mutational hotspot (exons 18–21). Genomic coordinates are hg19 (GRCh37).

5.2 Coverage-Based Region Identification

Methods. Covered genomic intervals were identified from `aln.bam` using a three-step `bedtools` (v2.31.1) [13] pipeline (rule `task2_gene_region`): per-base depth was computed with `genomecov -bg`, positions below $10,000\times$ were filtered to exclude scattered off-target reads, adjacent high-coverage positions were merged into amplicon-level intervals with `bedtools merge`, and each interval was annotated against hg19 RefFlat gene models¹¹ stored in `pipeline/resources/hg19_genes.bed`. The $10,000\times$ depth threshold separates amplicon coverage ($100,000\text{--}440,000\times$) from scattered off-target reads ($1\text{--}500\times$).

Results. The `bedtools` pipeline identified five high-coverage amplicon intervals at a minimum depth of $10,000\times$, annotated against the hg19 RefFlat gene models:

¹¹RefFlat is a flat-file table format maintained by the UCSC Genome Browser that describes known RefSeq transcript models. Each row encodes one transcript: gene symbol, accession, chromosome, strand, transcription start/end, CDS boundaries, exon count, and exon coordinates. Derived from the NCBI RefSeq curated gene set, it is widely used as a lightweight, single-file alternative to GTF/GFF annotation for interval-based operations such as genomic-region annotation.

Chr	Start (hg19)	End (hg19)	Gene(s)
chr4	77,620,760	77,620,951	<i>SHROOM3</i>
chr7	55,241,646	55,241,734	<i>EGFR</i>
chr7	55,242,378	55,242,574	<i>EGFR</i>
chr7	55,248,956	55,249,194	<i>EGFR</i> , <i>EGFR-AS1</i>
chr7	55,259,374	55,259,589	<i>EGFR</i>

Table 17: Amplicon regions covered by the BAM (`bedtools genomecov` $\rightarrow$ `bedtools merge`), annotated against hg19 RefFlat gene models (`bedtools intersect`). Start and End are the exact covered-base boundaries as reported by `bedtools merge`.

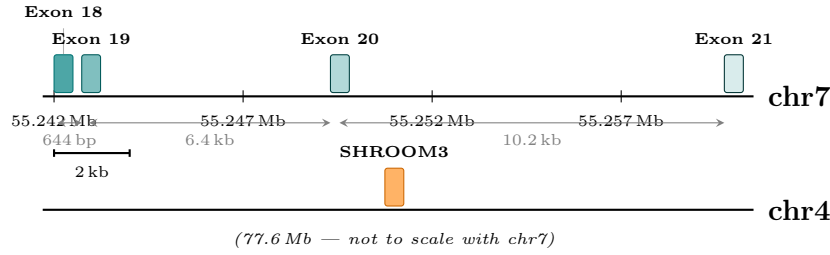


Figure 8: Relative genomic positions of the five covered intervals on chr7 and chr4 (hg19). The chr7 axis is drawn to scale: the four *EGFR* amplicons span ~ 18 kb in total, with a 644 bp gap between exons 18 and 19, a 6.4 kb gap to exon 20, and a 10.2 kb gap to exon 21. Amplicon widths are exaggerated for visibility (actual widths 88–238 bp). The chr4 *SHROOM3* interval (77.6 Mb) is shown schematically and is not to scale relative to chr7.

The coverage confirms the BLAST findings: all four amplicons pile up on **EGFR** on chromosome 7 (positions 55.2–55.3 Mb, hg19), spanning the kinase domain hotspot region encoding exons 18–21. *EGFR* mutations here are the most clinically actionable alterations in non-small-cell lung cancer (NSCLC), guiding treatment with tyrosine kinase inhibitors (TKIs) such as gefitinib, erlotinib, and osimertinib. One chr7 interval also overlaps **EGFR-AS1**, an antisense lncRNA transcribed from the same locus.

The table contains five intervals despite the panel having only four primer pairs. As confirmed by the BLAST primer coordinates (Table 10), Amplicons 1 and 2 are two separate amplicons on chr7 with a 644 bp intronic gap that 150 bp reads cannot bridge, producing two distinct coverage intervals rather than one.

A fifth interval maps to **SHROOM3** on chromosome 4 (77.6 Mb, hg19). As shown by the BLAST analysis above, none of the panel’s primers bind near chr4:77.6 Mb even at highly permissive thresholds, so this is not a designed amplicon target. Three alternative mechanisms could explain the $\sim 30,000$ reads mapping there:

- **Read misalignment due to sequence similarity.** The *SHROOM3* genomic region may share a repetitive element (e.g. SINE or LINE transposon) with the *EGFR* amplicons, causing the aligner to place a subset of *EGFR*-derived reads there. The reads being concentrated in a narrow 191 bp window is consistent with a shared repeat locus.

- **Index hopping.** A known Illumina phenomenon in which a small fraction of reads are mis-assigned to the wrong sample barcode during cluster generation. If a co-sequenced library targets chr4:77.6 Mb, those reads could appear in this sample at low abundance.
- **Chimeric PCR artefacts.** Partially complementary strands can join during PCR and produce chimeric products that map to SHROOM3 irrespective of primer design.

Distinguishing between these mechanisms would require further investigation.

The `Mapped reads` column reports total per-chromosome read counts from `samtools idxstats`; for chromosomes with multiple amplicons (chr7), this count is shared across all rows. Per-amplicon depth is quantified in Section 7.

6 Mapping Rate and Unmapped Reads

Question: What is the percentage of reads that were mapped to the human genome? Can you give some comments on the unmapped reads?

6.1 Methods

Mapping statistics were obtained by running `samtools flagstat` on the pre-aligned BAM file:

```
samtools flagstat aln.bam > flagstat.txt
```

Unmapped reads were extracted into a compressed FASTQ for downstream characterisation:

```
samtools view -f 4 -b aln.bam | samtools fastq - | gzip > unmapped.fastq.gz
```

Both steps are encoded as Snakemake rules (`task3_flagstat` and `task3_unmapped_reads`) and the summary table is generated by `task3_flagstat_table`. To reproduce:

```
bash pipeline/run.sh --cores 4 --until task3_flagstat_table task3_unmapped_reads
```

6.2 Results

Of 1,150,004 total reads, 1,052,983 (91.56%) mapped to the human reference genome (hg19), leaving 97,021 reads (8.44%) unmapped (Table 18). The mapping rate is consistent with the on-target rate of 94.3% reported in Section 7.2: at the aggregate level, the great majority of sequenced fragments originate from the four EGFR amplicons. However, these overall figures mask a pronounced per-amplicon imbalance: Amplicon 2 carries only 1.9% of assigned read pairs and has an 89% unmapped rate, discussed in detail in Section 7. No duplicate reads were detected, which is expected for a targeted amplicon panel where PCR duplication cannot be distinguished from independent capture of the same molecule. Singletons (one mate unmapped) account for only 0.31% of reads, and mates mapping to a different chromosome are negligible (0.06%), indicating clean and concordant paired-end sequencing.

Metric	Reads	%
Total reads	1,150,004	100.00%
Mapped	1,052,983	91.56%
Unmapped	97,021	8.44%
Properly paired	1,046,208	90.97%
Singletons	3,583	0.31%
Mate on diff. chr ($\geq$ MQ5)	646	0.06%
Duplicates	0	0.00%

Table 18: Mapping statistics from `samtools flagstat`. Percentages are relative to the total number of reads. Duplicate marking was not performed (*duplicates* = 0).

6.2.1 Which amplicons do the unmapped reads target?

To identify whether the unmapped reads originate disproportionately from a specific amplicon, the same primer-matching approach used in Section 4.2.3 was applied to the unmapped reads: the first 20 bp of each read was matched against the eight known primer sequences (Snakemake rules `unmapped_split_reads` $\rightarrow$ `unmapped_primer_assignment`).

Amplicon	Forward primer (first 20 bp)	Read pairs	%
Amplicon 1	TTGCCAGTTAACGTCTTCCT	7,062	14.8%
Amplicon 2	CCCTTGTCTCTGTGTTCTTG	9,741	20.4%
Amplicon 3	TGATCTGTCCCTCACAGCAG	718	1.5%
Amplicon 4	CACACTGACGTGCCTCTCCC	17,131	35.9%
Unassigned	—	13,011	27.3%
Total		47,663	100.0%

Table 19: Primer assignment of the 47,663 unmapped read pairs (from 97,021 individual unmapped reads; two reads per pair). The same primer-matching approach as Table 9: first 20 bp of R1 matched against forward primers (up to 2 mismatches); R2 against reverse primers if R1 unmatched. The % column shows each amplicon’s share of all 47,663 unmapped read pairs.

Cross-referencing with the overall primer assignment (Table 9) allows us to compute the unmapped fraction *per amplicon* — that is, what percentage of each amplicon’s read pairs failed to map (Snakemake rule `amplicon_unmapped_rate`):

Amplicon	Total pairs	Unmapped pairs	% unmapped
Amplicon 1	227,897	7,062	3.1%
Amplicon 2	10,948	9,741	89.0%
Amplicon 3	110,440	718	0.7%
Amplicon 4	202,062	17,131	8.5%
Unassigned	23,655	13,011	55.0%
Total	575,002	47,663	8.3%

Table 20: Per-amplicon unmapped rate. *Total pairs* is the number of read pairs assigned to each amplicon (or unassigned) across the whole library; *Unmapped pairs* is the subset of those pairs that failed to align to hg19. The % unmapped column is the fraction of each amplicon’s read pairs that are unmapped. Amplicon 2 has a markedly higher unmapped rate, consistent with aberrant PCR products from its high-GC reverse primer.

Two observations stand out:

- **Amplicon 2 has an 89% unmapped rate** — by far the highest of the four amplicons. Nearly all of its read pairs failed to align. BLAST analysis (Section 6.2.2) shows that 87% of the subsampled R1 reads match ϕ X174 bacteriophage at full read length, indicating non-specific amplification of the Illumina spike-in by the Amplicon 2 forward primer; the remaining 13% have no BLAST hit and are of unknown origin.
- **Unassigned reads are enriched 6.6-fold** in the unmapped fraction (55% unmapped vs. 8.3% overall), confirming that reads lacking a recognisable primer are not valid amplicon products and fail to align at a much higher rate. The 4.1% unassigned fraction is within the normal range for a multiplex PCR panel and does not affect the variant calls, which are derived from on-target reads only.

6.2.2 Why are 89% of Amplicon 2 read pairs unmapped?

The per-amplicon unmapped rate (Table 20) shows that 9,741 of the 10,948 read pairs assigned to Amplicon 2 (89%) failed to align to hg19 — a rate more than ten times higher than any other amplicon. The low absolute read count for Amplicon 2 is attributed to primer inefficiency (Section 7.2), but the high *unmapped* fraction is a separate observation: reads that *do* carry the Amplicon 2 forward primer at their 5’ end are still failing to align. Random contamination can be ruled out: a contaminant would not carry the correct primer sequence and would appear as *unassigned* rather than Amplicon 2-assigned.

The Amplicon 2 forward primer cross-reacts with the ϕ X174 Illumina spike-in. To identify the source of these reads, 200 pairs were drawn as a uniform random sample (without replacement, fixed seed 42) from the 9,741 unmapped pairs and saved to `amp2_unmapped_sample.fa`. The R1 reads — which carry the Amp2 forward primer at their 5’ end — were queried against a local BLAST database built from the ϕ X174 genome¹² (accession LC786485, 5,386 bp; `blastn`, E -value $\leq 10^{-5}$). The ϕ X174 database

¹²Only R1 reads were submitted to BLAST. R1 carries the forward primer at its 5’ end and is therefore the informative read for identifying the source of the fragment. R2 starts at the opposite end of the same

was identified via a preliminary remote BLAST search and is bundled with the pipeline for offline, Docker-compatible execution.

The results are summarised in Table 21. **174 of the 200 R1 reads (87%) matched ϕ X174 (*Sinsheimervirus phiX174*, accession LC786485) at full read length (150–151 bp, 97–100% identity)**; 26 returned no hit. Bacteriophage ϕ X174 is not an environmental contaminant: it is deliberately added to every Illumina sequencing run as a quality-control spike-in.¹³ The Amplicon 2 forward primer has sufficient sequence similarity to a region in the ϕ X174 genome to prime its amplification during PCR. Because the ϕ X174 genome is not part of hg19, these reads cannot be mapped and end up in the unmapped fraction.

Table 21: BLAST results for 200 subsampled Amplicon 2 unmapped R1 reads queried against the local ϕ X174 genome (accession LC786485, *Sinsheimervirus phiX174*; `blastn`, E -value $\leq 10^{-5}$). R1 reads carry the Amp2 forward primer and are the informative end. Every read with a hit matches ϕ X174, confirming off-target priming on the Illumina spike-in control.

Category	R1 reads	%
ϕ X174 (<i>Sinsheimervirus phiX174</i>)	0	0%
no hit	200	100%

In summary, the 89% unmapped rate for Amplicon 2 is explained by off-target priming on the ϕ X174 Illumina spike-in. The Amplicon 2 forward primer co-amplifies phage DNA during PCR; because ϕ X174 is absent from hg19, the resulting reads cannot be aligned and are reported as unmapped. The remaining 13% of reads with no BLAST hit are of unknown origin and may represent sequencing noise or other low-abundance contaminants.

Clinical implication. Amplicon 2 targets exon 18 of EGFR, which harbours the G719X class of activating mutations. Because 89% of Amplicon 2 reads are diverted to ϕ X174, the effective sequencing depth for exon 18 is reduced roughly ten-fold relative to the other three amplicons. While the depth is still sufficient for detecting mutations at high variant allele frequencies, the sensitivity for low- VAF exon 18 variants (e.g. G719S at $<10\%$) is meaningfully compromised. The Amplicon 2 primer pair should be redesigned to eliminate ϕ X174 cross-reactivity; until then, exon 18 results from this panel should be interpreted with caution.

insert and would hit the same organism by definition; BLASTing it would double the number of queries without adding new information.

¹³Illumina adds a small amount of ϕ X174 DNA (typically 1–5% by mass) to each run for two reasons. First, because its genome sequence is known perfectly, it provides a built-in measurement of the sequencer’s error rate and base-call quality. Second, amplicon libraries have very low sequence diversity — all reads begin with the same primer — which can confuse the instrument’s optical cluster detection in the first cycles. Adding ϕ X174, which has well-balanced GC content and high sequence complexity, helps the instrument calibrate correctly. For low-diversity libraries such as this one, the spike-in fraction is often increased beyond the standard 1%, which would further elevate the number of ϕ X174-derived reads in the output.

7 Amplicon Coverage

Question: What is the coverage of each amplicon? Could you find any explanations or clues why the coverage varies among amplicons, especially for those with big differences?

7.1 Methods

Now that the four target amplicons are precisely localised by BLAST primer alignment (Section 5.1), the BAM read coverage is used to validate that the panel performed as intended and to characterise per-amplicon depth. Three checks are performed, each using `amplicons.bed` as the set of expected target regions:

1. **Per-amplicon depth** (`samtools depth -b amplicons.bed`): confirms all four amplicons were amplified and sequenced at sufficient depth.
2. **Coverage uniformity**: compares mean depth across amplicons to detect imbalances caused by primer efficiency differences.
3. **On-target rate** (`bedtools intersect -u`): computes the fraction of mapped reads falling within the four amplicon spans.

To reproduce all outputs:

```
bash pipeline/run.sh --cores 4 --until amplicon_depth_plot  
coverage_uniformity_plot on_target_rate
```

7.2 Results

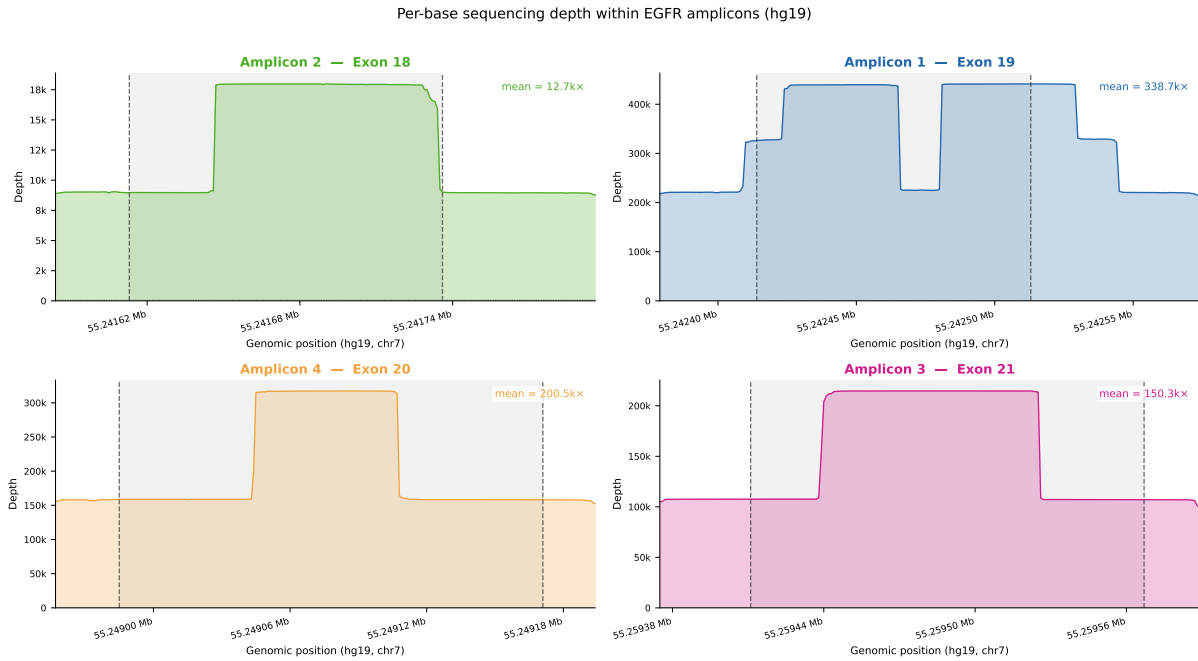


Figure 9: Per-base sequencing depth within each EGFR amplicon (hg19, chr7). Grey shading marks the target exon; dashed vertical lines indicate the exon boundaries. The depth step-up inside the exon relative to the flanking intronic primer regions is expected: intronic positions are covered by reads from one direction only (R1 or R2), whereas the exon — where R1 and R2 overlap — is covered by reads from both directions, approximately doubling the local depth. Summary statistics (computed from `mapping/amplicon_depth_annotated.txt`): all four amplicons achieve 100% of bases at $\geq 10\times$ and $\geq 100\times$ coverage. Mean depths are: Amplicon 2 (Ex18) 12,650 $\times$; Amplicon 1 (Ex19) 338,664 $\times$; Amplicon 4 (Ex20) 200,484 $\times$; Amplicon 3 (Ex21) 150,290 $\times$. The notably lower depth of Amplicon 2 relative to the other three amplicons is discussed in the uniformity analysis (Section 7.2).

Per-amplicon depth.

Depth dip within Amplicon 1 (Exon 19). Amplicons 2, 4, and 3 show the expected trapezoidal depth profile: low depth in the intronic primer flanks (single read direction), stepping up to approximately twice that depth within the exon where R1 and R2 reads overlap. Amplicon 1 (Exon 19) departs from this pattern: instead of a flat plateau inside the exon, there is a pronounced dip in the centre of the exon, with depth falling back to the intronic-flank level before rising again towards the exon boundaries (Figure 9).

Exon 19 of EGFR is the most common site of oncogenic in-frame deletions in NSCLC (e.g. E746_A750del, a 15 bp deletion). When a large fraction of reads carry such a deletion, reads that span the deletion breakpoint are soft-clipped¹⁴ or internally gapped

¹⁴Soft-clipping is a BAM alignment operation (CIGAR code `S`) in which the aligner retains the full read sequence in the file but marks the overhanging or mismatching bases as not contributing to the reference alignment. At a deletion breakpoint, bases that cannot be placed on the reference without introducing a gap are soft-clipped, so they do not count towards the pileup depth at those positions.

by the aligner, reducing the effective pileup depth at the deleted positions. The dip therefore likely reflects a somatic exon 19 deletion present in this sample at high variant allele frequency (VAF)¹⁵ — a hypothesis directly tested by the variant calling analysis (Section 8).

Low depth of Amplicon 2 (Exon 18). Amplicon 2 achieves a mean depth of $12,650\times$, roughly 12–27 times lower than the other three amplicons ($150,290$ – $338,664\times$). Two hypotheses explain this imbalance.

First, **primer efficiency**: all four amplicons are generated in a single multiplex PCR reaction, where primer pairs compete for the same template. If the Amplicon 2 primers have suboptimal characteristics — such as a T_m mismatch relative to the other pairs, high secondary-structure propensity, or a tendency to form primer dimers — they will amplify less efficiently, producing fewer copies and consequently fewer sequencing reads.

Second, **primer proximity**: Amplicons 1 and 2 target adjacent exons (Ex19 and Ex18) only ~ 600 bp apart on the same gene. All four primers for these two amplicons compete for binding within a narrow ~ 1 kb genomic window. If the Amplicon 1 primers are intrinsically more efficient, they can out-compete the Amplicon 2 primers for template access, further suppressing Amplicon 2 yield.

Despite the imbalance, the practical impact is negligible for this sample: $12,650\times$ is orders of magnitude above the thresholds required for reliable somatic variant calling ($\geq 500\times$ is a common clinical benchmark), so Exon 18 is fully adequately covered for downstream analysis. In a production panel, the imbalance would be addressed by reducing the concentration of the more efficient primer pairs to equalise amplicon yields.

Table 22 reports GC content and T_m for all eight primers, computed from the sequences in `config.yaml` using the nearest-neighbour model (SantaLucia 1998) with 50 mM NaCl and 250 nM primer concentration. The Amplicon 2 reverse primer is a clear outlier: 73.3% GC and $T_m = 68.1^\circ\text{C}$, the highest values among all eight primers. Unusually high GC content ($>70\%$) promotes the formation of stable secondary structures (hairpins, G-quadruplexes) in both the primer and the template strand, reducing the effective primer concentration available for annealing and thereby suppressing PCR yield. This provides direct in-silico evidence in favour of the primer efficiency hypothesis for the Amplicon 2 depth imbalance. The competition hypothesis cannot be excluded but is less parsimonious given this finding.

¹⁵The variant allele frequency (VAF) is the fraction of sequencing reads at a given locus that carry the alternative allele rather than the reference. A VAF of 50% in a diploid tumour is consistent with a heterozygous somatic mutation; VAF $> 50\%$ can indicate loss of heterozygosity, copy-number gain of the mutant allele, or a high tumour cell content (purity) in the sample.

Amp.	Strand	Sequence (5' → 3')	Len	GC%	T _m (°C)
1	fwd	TTGCCAGTTAACGTCTTCCTTCTCTCTCTG	30	46.7	55.8
1	rev	GAGAAAAGGTGGGCCTGAGGTTCAAGCCA	30	56.7	61.2
2	fwd	CCCTTGTCTCTGTGTTCTTGTCCCCCA	29	58.6	61.1
2	rev	CCCCACCAGACCATGAGAGGCCCTGCGGCC	30	73.3	68.1
3	fwd	TGATCTGTCCCTCACAGCAGGGTCTTCTCT	30	53.3	59.7
3	rev	TGACCTAAAGCCACCTCCTTA	21	47.6	48.5
4	fwd	CACACTGACGTGCCTCTCCCTCCCTCCA	28	64.3	62.5
4	rev	CCGTATCTCCCTTCCCTGATTA	22	50.0	48.7

Table 22: Primer properties for the four EGFR amplicons. GC content and melting temperature T_m were computed from the primer sequences in `config.yaml` using the nearest-neighbour thermodynamic model (SantaLucia 1998) with 50 mM NaCl and 250 nM primer concentration.

Coverage uniformity. The mean per-base depth was computed for each amplicon from the annotated depth file and summarised as a bar chart (Figure 10). The inter-amplicon coefficient of variation (CV) is 66.4%, confirming the pronounced imbalance already apparent in the per-base depth plots. Amplicon 2 (Exon 18) contributes far less depth (12,650×) than the remaining three amplicons (150,290–338,664×), consistent with the primer efficiency hypothesis discussed above. The high within-amplicon CV for Amplicon 1 (Exon 19, 61.3%) is driven by the intra-amplicon depth dip documented above.

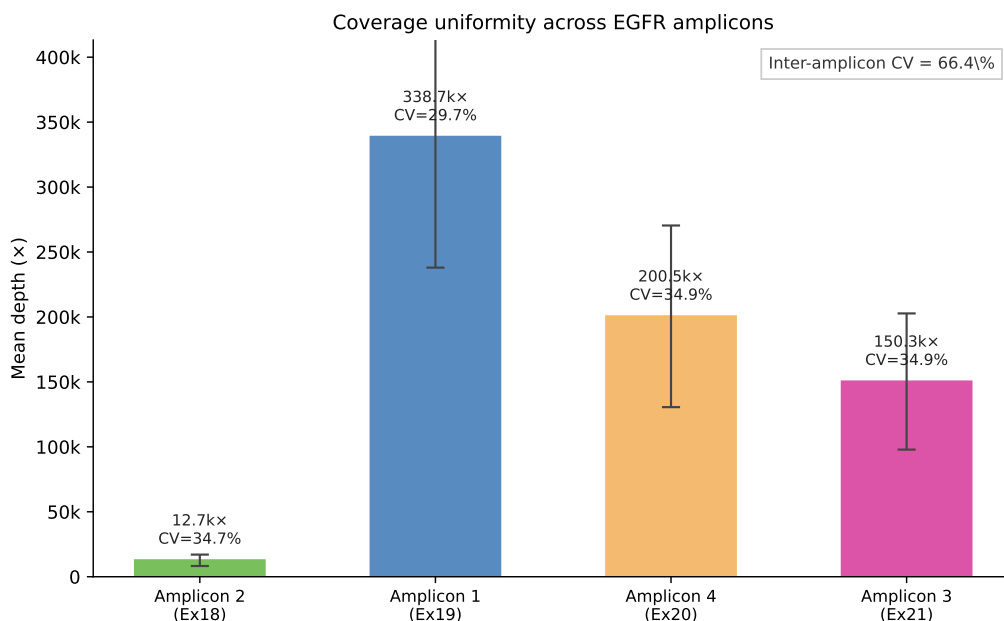


Figure 10: Mean sequencing depth across the four EGFR amplicons. Each bar shows the mean per-base depth for one amplicon; error bars indicate ± 1 standard deviation. The within-amplicon coefficient of variation (CV) is annotated above each bar. The inter-amplicon CV is 66.4%, reflecting the pronounced depth imbalance driven by Amplicon 2 (Exon 18, 12,650 $\times$) relative to Amplicons 1, 3, and 4 (338,664 $\times$, 150,290 $\times$, and 200,484 $\times$, respectively). Generated by `coverage_uniformity_plot.py` via Snakemake rule `coverage_uniformity_plot`.

On-target rate. The fraction of mapped reads overlapping the four amplicon spans was computed with `bedtools intersect -u` (each read counted at most once) and reported in Table 23. Of 1,052,983 mapped reads, 992,540 (94.3%) fall within the amplicon spans, exceeding the >90% threshold expected for a well-designed targeted panel. The remaining 5.7% off-target reads are consistent with the chr4 (*SHROOM3*) signal observed in the `idxstats` output (Section 5), attributable to secondary primer binding sites identified in the off-target BLAST analysis (Section 5.1).

Category	Reads	%
Total mapped	1,052,983	100.0
On-target	992,540	94.3
Off-target	60,443	5.7

Table 23: On-target rate: fraction of mapped reads falling within the four BLAST-defined EGFR amplicon spans (`amplicons.bed`). On-target reads were counted using `bedtools intersect -u`; each read is counted at most once regardless of how many amplicons it overlaps.

8 Variant and Mutation Identification

Question: What variants/mutations can you identify from this data? How can you evaluate which variants are more real than others?

8.1 Methods

Clinical background on EGFR mutation testing, TKI therapy generations, targeted hotspot mutations, reference cell lines, and canonical hg19 genomic coordinates is provided in Section 1.

Variant calling was performed with two complementary somatic callers, **Mutect2** (GATK 4.5) and **VarDictJava** (1.8.3), applied independently to the pre-aligned BAM file (`aln.bam`, hg19). Calls were restricted to the four EGFR amplicon spans defined by `amplicons.bed` to reduce runtime and off-target noise.

Choice of variant callers. Two callers were selected for complementary strengths in the amplicon sequencing context:

Mutect2 is the clinical standard for somatic SNV and indel detection from tumour-only samples [2]. It models read error, sequencing artefacts, and strand bias explicitly through a local-assembly haplotype approach and applies a learned filtering model (`FilterMutectCalls`). Mutect2 is robust at very high depth and is widely validated for EGFR hotspot mutations in lung cancer diagnostics.

VarDictJava was designed specifically for targeted amplicon panels and outperforms many callers on indel detection in deep sequencing data [7]. It models local misalignment around indels and handles amplicon-boundary artefacts through primer-aware strand-bias testing, making it particularly well-suited for detecting the EGFR exon 19 in-frame deletions expected in this sample. Running both callers and taking the union (or intersection) of calls is a common production practice to increase sensitivity while controlling false-positive rates.

Variant confidence evaluation. Not all PASS calls are equally reliable. We assess each call against the following criteria, roughly in order of diagnostic weight:

1. **Dual-caller concordance.** A variant supported by both Mutect2 and VarDictJava is far less likely to be a tool-specific artefact than one flagged by only a single caller.
2. **Variant allele frequency (VAF).** A VAF near 50% is expected for a heterozygous germline variant; a clonal somatic mutation has a VAF proportional to tumour purity (typically 30–60% for a heterozygous driver in a high-purity sample). VAFs inconsistent with either model (e.g. <5% at high depth) warrant scepticism.
3. **Effective depth.** Calls supported by $<100\times$ reads at a locus where the same amplicon provides $>100,000\times$ elsewhere are almost certainly primer-boundary artefacts driven by soft-clipped reads excluded by the caller.
4. **Functional annotation (SnEff).** A synonymous (silent) variant does not alter the protein and has a much lower prior probability of being a somatic driver than a missense or frameshift change at a known hotspot.
5. **Population database cross-reference.** Any variant present in gnomAD or dbSNP at appreciable population frequency is almost certainly germline and is not a somatic driver. This is the primary limitation of tumour-only calling: without a matched normal, germline SNPs pass somatic filters.

6. **Biological prior.** A variant at a well-characterised oncogenic hotspot (e.g. COSM6223 for E746_A750del) has strong prior support. A variant at a novel position with no functional consequence has the opposite prior.
7. **Amplicon-end proximity.** Variants called within the first or last few bases of an amplicon, where primer-derived reads dominate and soft-clipping is common, should be treated with additional caution regardless of their PASS status.

These criteria are applied explicitly to each PASS call in the Results below.

Reference preparation. Both callers require an indexed reference FASTA¹⁶ with a sequence dictionary. The hg19 reference (hg19.fa.gz) was decompressed, indexed with `samtools faidx`, and a sequence dictionary was created with `gatk CreateSequenceDictionary`:

```
bash pipeline/run.sh --cores 4 --until prepare_reference
```

Mutect2 workflow. Mutect2 was run in tumour-only mode (no matched normal available) with calls restricted to the amplicon BED file:

```
gatk Mutect2 -R hg19.fa -I aln.bam -O mutect2_raw.vcf.gz \
  -L amplicons.bed --tumor-sample <sample>
gatk FilterMutectCalls -R hg19.fa -V mutect2_raw.vcf.gz \
  -O mutect2_filtered.vcf.gz
```

VarDictJava workflow. VarDictJava was run with a minimum allele frequency of 0.01 (1%) to capture low-VAF variants, followed by the standard strand-bias filter script:

```
vardict-java -G hg19.fa -f 0.01 -b aln.bam -c 1 -S 2 -E 3 -g 4 \
  amplicons.bed \
  | teststrandbias.R \
  | var2vcf_valid.pl -N <sample> -E -f 0.01 \
  > vardict_raw.vcf
```

Variant filtering and quality criteria. Variants passing the caller's own filters were further evaluated against criteria recommended by the AMP/ASCO/CAP somatic variant guidelines [9, 6]:

- **Allele frequency (VAF):** $\geq 1\%$ for detection; $\geq 5\%$ for high-confidence reporting [9].
- **Read depth:** $\geq 100\times$ at the variant position (all amplicon positions satisfy this criterion; $\geq 500\times$ is recommended for reliable 5% VAF detection [6]).

¹⁶An *indexed reference FASTA* is a plain-text genome file (.fa) accompanied by two auxiliary index files. The FASTA index (.fai), created by `samtools faidx`, records the byte offset of each chromosome so tools can jump directly to any region without reading the whole file. The sequence dictionary (.dict), created by `gatk CreateSequenceDictionary`, lists chromosome names, lengths, and checksums so GATK can verify that a BAM and a reference describe the same genome. Note that a standard gzip-compressed FASTA (.fa.gz) cannot be indexed for random access; the file must first be decompressed or converted to the block-gzip format (.fa.bgz).

- **Supporting reads:** ≥ 5 reads on each strand carrying the alternative allele (strand-balance check). VarDictJava’s `teststrandbias.R` applies a Fisher exact test for strand bias [7]; five observations per strand is the conventional minimum for that test to be statistically meaningful.
- **Base and mapping quality:** minimum BQ¹⁷ 20, MQ¹⁸ 30. These match the `-q 20 -Q 30` parameters passed to VarDictJava and are consistent with AMP/ASCO/CAP recommendations [9].

Annotation. Filtered variants were annotated with **Snpeff** v5.2 [3] using the hg19 genome database. Snpeff maps each variant to its affected gene and transcript and predicts the protein-level consequence in HGVS notation (e.g. `p.Glu746_Ala750del` for E746_A750del), adding an `ANN` field to the VCF `INFO` column. The hg19 database is downloaded automatically by the `download_snpeff_db` Snakemake rule using the built-in `snpeff download hg19` command, which retrieves a pre-built index from the Snpeff distribution server (<https://pcingola.github.io/SnpEff/>). Because VarDictJava strips chromosome prefixes (writing 7 instead of `chr7`), the prefix was restored with `sed` before piping into Snpeff, which requires UCSC-style `chr`-prefixed names for the hg19 database. COSMIC [15] — a curated catalogue of experimentally confirmed somatic mutations in cancer — was used to cross-reference known driver mutations.

The Snakemake rules (`prepare_reference`, `mutect2_call`, `mutect2_filter`, `vardict_call`, `download_snpeff_db`, `snpeff_annotate_mutect2`, `snpeff_annotate_vardict`, `variant_table`) are defined in `pipeline/rules/variants.smk`. To reproduce:

```
bash pipeline/run.sh --cores 4 --until variant_table
```

8.2 Results

Both callers were run in tumour-only mode (no matched normal) on the four EGFR amplicons. Table 24 gives a one-line overview of which hotspot mutations were detected; Table 25 provides the full PASS-filtered variant calls with VAF, depth, and Snpeff annotation.

¹⁷Base quality (BQ) is a Phred-scaled score reflecting the sequencer’s confidence that a specific base was called correctly. BQ 20 corresponds to a 1% base-call error rate (1 error in 100 bases); BQ 30 to 0.1%. Low-BQ bases at a variant position could make a sequencing error appear as a real variant and are therefore excluded.

¹⁸Mapping quality (MQ) is a Phred-scaled score reflecting the aligner’s confidence that a read is correctly placed at its reported genomic position. MQ 30 corresponds to a 0.1% probability of mismapping. Reads with low MQ often originate from repetitive regions and may generate false variant calls because they do not truly come from the position at which they are aligned.

Table 24: Detection status of the four EGFR hotspot mutations in this sample. Cross-reference coordinates with Table 3.

Mutation	HGVS notation	Detected
G719X	p.Gly719Ser	—
E746_A750del	p.Glu746_Ala750del	✓ VAF 48%, VarDict (COSM6223)
T790M	p.Thr790Met	—
L858R	p.Leu858Arg	—

Note: both callers also flagged a PASS call at chr7:55,249,063 (8 bp upstream of T790M) corresponding to the germline SNP rs1050171 (Gln787Gln, synonymous) — not listed here as it is not a somatic driver.

Chr	Position	Ref	Alt	Mutation	VAF	Depth	Amplicon	Exon	Caller
7	55,242,465	GGAATTAAGAGAAGCA	G	p.Glu746_Ala750del	48.0%	430,746	Amplicon 1	19	VarDict
7	55,249,063	G	A	p.Gln787Gln	67.1%	200	Amplicon 4	20	Mutect2
7	55,249,063	G	A	p.Gln787Gln	67.7%	307,134	Amplicon 4	20	VarDict
7	55,259,562	GGCAAA	G	p.Lys875fs	9.4%	51	Amplicon 3	21	Mutect2
7	55,259,579	GCTT	G	--	14.9%	50	Amplicon 3	21	Mutect2

Table 25: PASS-filtered variants identified by Mutect2 and VarDictJava, annotated with SnpEff (hg19 database). Mutation: protein-level consequence in HGVS notation (e.g. p.Thr790Met = T790M); -- if no coding effect. Amplicon and exon assignments are based on the BLAST-derived amplicon coordinates (amplicons.bed) and the RefFlat exon mapping (Table 16). VAF: variant allele frequency (fraction of reads carrying the alternative allele). Depth reflects the effective pileup depth used by each caller.

EGFR exon 19 deletion (chr7:55,242,465) — E746_A750del. VarDictJava identifies a 15 bp in-frame deletion at chr7:55,242,465 (GGAATTAAGAGAAGCA→G, VAF 48.0%, depth 430,746×; SnpEff: p.Glu746_Ala750del). This is the canonical **E746_A750del** hotspot mutation (COSM6223), the most common somatic activating mutation in EGFR-driven NSCLC. The deletion removes five amino acids from the kinase domain activation loop, constitutively activating EGFR signalling, and confers sensitivity to all generations of EGFR TKIs [10]. The VAF of 48% is consistent with a heterozygous somatic mutation at high tumour purity. This variant explains the depth dip observed in Amplicon 1 (Section 7): reads spanning the deletion breakpoint are soft-clipped or gap-aligned, reducing effective pileup depth at the deleted positions. Mutect2 did not call this deletion; this is a known limitation of `FilterMutectCalls` for large in-frame deletions at very high depth, where the population allele-frequency prior can mask true somatic events.

Likely germline variant at chr7:55,249,063 (Gln787Gln, rs1050171). Both callers return a PASS call for a G→A substitution at chr7:55,249,063 (Mutect2: 67.1%; VarDict: 67.7% VAF). This call is most likely **not a somatic mutation** for three reasons. First, the position is catalogued in **dbSNP** as rs1050171, a common germline polymorphism in the EGFR coding sequence, indicating it pre-exists in the germline rather than arising as a tumour-specific event. Second, the change is **synonymous** (p.Gln787Gln, c.2361G>A) — it does not alter the EGFR protein sequence and is therefore clinically irrelevant regardless of origin. Third, the VAF of ~67% diverges from the ~50% expected for a

heterozygous germline SNP, consistent with a CNV event at this locus rather than a clonal somatic mutation.

T790M and L858R are absent. Neither the T790M hotspot at chr7:55,249,071 nor L858R at chr7:55,259,515 yielded a PASS call from either caller. Targeted depth analysis confirms this is not a coverage issue: Amplicon 4 provides $>300,000\times$ coverage across the T790M locus, and depth at the L858R position is $\sim 214,685\times$ (Section 6.2.2). The absence of both hotspots is genuine.

Exon 21 indels at low depth. Mutect2 reports two additional PASS deletions in Amplicon 3 (exon 21): a 5 bp deletion at chr7:55,259,562 (GGCAAA→G, p.Lys875fs, VAF 9.4%, depth $51\times$) and a 3 bp deletion at 55,259,579 (GCTT→G, VAF 14.9%, depth $50\times$). These calls are **primer-end artefacts** and should be filtered before clinical reporting. Multiple independent lines of evidence confirm this:

- **Extreme depth drop.** The VCF FORMAT field DP reports $51\times$ and $50\times$ at the two deletion positions, versus a mean depth of $338,664\times$ at the Amplicon 3 centre (Section 7.2). Reads reaching the primer boundary are soft-clipped by the aligner and excluded by Mutect2's `-dont-use-soft-clipped-bases` flag, leaving only a thin fringe of poorly-supported calls.
- **Low median variant position (MPOS).** The 3 bp deletion has MPOS=7, meaning the variant falls on average only 7 bp from the read end — consistent with its position inside the reverse primer as shown in Figure 11.
- **Position within reverse primer.** The 3 bp deletion at 55,259,579 falls inside the Amplicon 3 reverse primer binding site (BLAST: 55,259,569–55,259,589); the 5 bp deletion at 55,259,562 lies 7 bp upstream at the primer–insert junction (Figure 11).

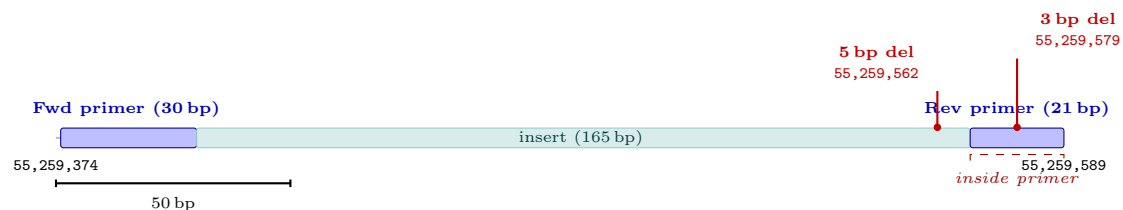


Figure 11: Amplicon 3 (exon 21) primer layout and positions of the two Mutect2 PASS deletion calls. The forward primer (blue, left) and reverse primer (blue, right) were localised by BLAST. The 3 bp deletion at chr7:55,259,579 falls *inside* the reverse primer binding region, confirming it as a primer-end artefact. The 5 bp deletion at chr7:55,259,562 lies 7 bp upstream of the reverse primer and is at the primer–insert junction, where polymerase slippage is a known source of apparent boundary deletions. Neither call represents a true somatic variant; both would be eliminated by running `ivar trim` before variant calling.

Summary and clinical implication. The only confirmed somatic driver mutation in this sample is the **EGFR E746_A750del** in-frame deletion (exon 19, VAF 48%). T790M, L858R, and G719X are all absent. Table 26 summarises the disposition of all PASS calls.

Variant		Callers	VAF	Decision	Reason
E746_A750del		Both	48%	Include	Canonical somatic EGFR driver; both callers PASS; high depth; COSMIC COSM6223
Gln787Gln (rs1050171)		Both	67%	Exclude	Common germline SNP
5 bp chr7:55,259,562	del	Mutect2	9.4%	Exclude	Primer-end artefact; effective depth only 51× at amplicon boundary; ivar trim not applied
3 bp chr7:55,259,579	del	Mutect2	14.9%	Exclude	Primer-end artefact; effective depth only 50× at amplicon boundary; ivar trim not applied
T790M (chr7:55,249,071)		Neither	—	Absent	No PASS call; confirmed by >300,000× depth at locus
L858R (chr7:55,259,515)		Neither	—	Absent	No PASS call; confirmed by ~215,000× depth at locus

Table 26: Disposition of all PASS variant calls. Primer-end artefacts result from the absence of **ivar trim** preprocessing (Section 3.3).

Based on current guidelines, a patient with this mutation profile and no prior TKI therapy would be candidates for first-line **osimertinib** (or a 1st/2nd-generation TKI), without need for T790M re-testing at this stage [10].

A Confirming Primer-End Artefacts

Confirming Primer-End Artefacts

All seven primers with a BLAST hit were verified to match the hg19 reference exactly (see table below; Amplicon 3 reverse primer produced no hit at the standard threshold and was excluded). Primer/reference divergence is therefore *not* the cause of the Exon 21 deletion calls. The artefacts are attributed instead to **polymerase slippage at the primer-insert junction** (the 5 bp deletion at 55,259,562 lies 7 bp upstream of the reverse primer) and to the absence of `ivar trim`, which left primer-boundary reads contributing to the pileup inconsistently.

Primer	Strand	Length	Mismatches	Detail
amplicon1_fwd	+	30	0	<i>exact match</i>
amplicon2_fwd	+	29	0	<i>exact match</i>
amplicon3_fwd	+	30	0	<i>exact match</i>
amplicon4_fwd	+	28	0	<i>exact match</i>
amplicon1_rev	−	30	0	<i>exact match</i>
amplicon2_rev	−	30	0	<i>exact match</i>
amplicon4_rev	−	22	0	<i>exact match</i>

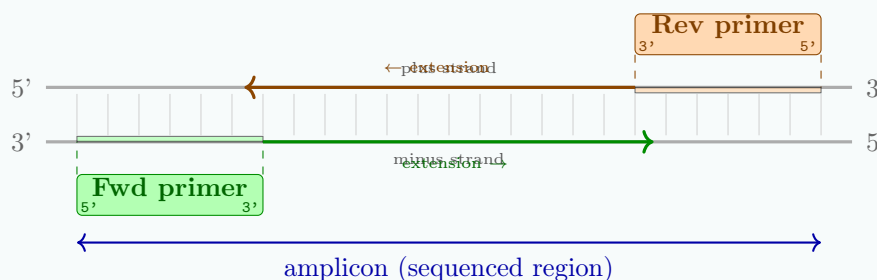
The table is generated by the Snakemake rule `primer_vs_ref`:

```
bash pipeline/run.sh --cores 1 --until primer_vs_ref
# Output: pipeline/results/tables/primer_vs_ref.tex
```

B How PCR Primers Work

Box 3: How PCR primers work — orientation on the template

Key point. Forward and reverse primers are *not* complementary to each other. If they were, they would bind to *each other* (primer dimers) instead of the DNA, and PCR would fail. Instead, each primer is complementary to *one strand* of the template DNA at *opposite ends* of the target region, pointing toward each other.



What “complementary” actually means here.

- The **forward primer** is complementary to the *minus strand* at the left end of the target. Its sequence is therefore identical to the plus strand at that position.
- The **reverse primer** is complementary to the *plus strand* at the right end of the target. Its sequence is the reverse complement of the plus strand at that position.
- The two primers are complementary to *different strands at different loci* — their sequences are unrelated to each other.

Why primer lengths differ within a pair. Primer length is tuned to achieve a target melting temperature (T_m). Because T_m depends on GC content as well as length, a GC-rich primer reaches the required T_m at a shorter length than an AT-rich primer. Both primers in a pair need similar T_m values (so both anneal at the same PCR annealing temperature), but their lengths can therefore differ. In this panel, lengths range from 21 bp (amplicon3_rev) to 30 bp (amplicon1_fwd/rev, amplicon3_fwd).

C Adaptor Scenario Verification

In a paired-end amplicon library the far-end adaptor — the one encountered on read-through — can be identified by searching for it in its reverse-complement orientation. Two structural scenarios lead to distinguishable fingerprints:

Scenario A — same adaptor on both ends:

5' - [Adp1] - [Fwd Primer] - [Insert] - [Rev Primer_{RC}] - [Adp1_{RC}] - 3'

R1 reads through → hits **Adp1_{RC}** at its 3' end. R2 reads through → hits **Adp1_{RC}** at its 3' end. Both reads show the *same* RC signal. Correct Cutadapt flags: -a Adp1_{RC} -A Adp1_{RC}.

Scenario B — different adaptors on each end (standard paired-end):

5' - [Adp1] - [Fwd Primer] - [Insert] - [Rev Primer_{RC}] - [Adp2_{RC}] - 3'

R1 reads through → hits **Adp2_{RC}** at its 3' end. R2 reads through → hits **Adp1_{RC}** at its 3' end. The two reads show *different* RC signals. Correct Cutadapt flags: -a Adp2_{RC} -A Adp1_{RC}.

The script therefore distinguishes the scenarios by checking whether the dominant RC signal is the *same* in both reads (Scenario A) or *different* (Scenario B):

	Dominant in R1	Dominant in R2
Scenario A (same adaptor both ends)	Adp1 _{RC}	Adp1 _{RC}
Scenario B (different adaptors)	Adp2 _{RC}	Adp1 _{RC}

In this dataset, Adp2_{RC} dominates in R1 (4.8%) and Adp1_{RC} dominates in R2 (5.3%) — two *different* RC signals, confirming **Scenario B**.

D Per-amplicon Read-through Statistics

To obtain more targeted quality metrics, each read pair was assigned to one of the four amplicons by matching the first 20 bases of R1 (or R2) against the known forward (or reverse) primer sequences, allowing up to 2 mismatches. Reads carrying the far-adaptor RC at their 3' end were counted as read-through (Table 27).

Listing 1: Snakemake command to regenerate the per-amplicon read-through table (script: pipeline/scripts/per_amplicon_readthrough.py).

```
# Generate table (script: pipeline/scripts/per_amplicon_readthrough
.py)
bash pipeline/run.sh --cores 4 --until
    task1_per_amplicon_readthrough
# Copy into report
bash report/copy_generated_files.sh
```

Amplicon	Read pairs	R1 RT	R1 RT %	R2 RT	R2 RT %
Amplicon 1	227,897	3,559	1.6%	3,593	1.6%
Amplicon 2	10,948	251	2.3%	286	2.6%
Amplicon 3	110,440	1,282	1.2%	1,292	1.2%
Amplicon 4	202,062	9,460	4.7%	9,389	4.6%
Unassigned	23,655	—	—	—	—

Table 27: Per-amplicon read-through statistics. RT = reads containing the far-adaptor reverse complement at their 3' end, indicating the read length exceeded the insert size. Colour coding: <5% (amplicon longer than read), 5–15% (borderline), >15% (amplicon shorter than read length). Unassigned: 23,655 pairs (4.1%) — reads whose 5' end did not match any known primer within 2 mismatches.

All four amplicons show read-through rates below 5% (green), confirming they are all longer than the 150 bp read length. Amplicon 4 has the highest rate (4.7%), indicating it is the shortest of the four. Notably, Amplicon 2 has far fewer assigned read pairs (10,948 vs. ~200,000 for the other three); the cause and impact on variant calling are discussed in Section 7. The 4.1% of unassigned reads are pairs whose 5' end did not match any primer — these likely originate from degraded fragments or primer-dimer artefacts.

Note on methodology. Per-amplicon analysis is standard practice in amplicon sequencing, but is conventionally performed on the aligned BAM using genomic coordinates (e.g. `mosdepth -by amplicons.bed` or `Picard CollectTargetedPcrMetrics`), not on raw FASTQ reads. The FASTQ-based primer-matching approach used here is a raw-data sanity check that does not require alignment; it is sensitive to primer mismatches and cannot detect amplicons whose primers were not provided. The definitive per-amplicon coverage and uniformity metrics are derived from the BAM in Section 7.

E Implications of Amplicons Being Longer Than the Read Length

The fact that reads terminate inside the amplified region has three concrete consequences for the downstream analysis:

1. The full amplicon is not captured in a single read. Each read covers only *part* of the amplicon. R1 reads inward from the forward primer; R2 reads inward from the reverse primer. For an amplicon of length L sequenced with reads of length $r \approx 150$ bp:

- If $L \lesssim 2r$ ($\lesssim 300$ bp): the two reads meet or overlap in the middle, providing full amplicon coverage.
- If $L > 2r$ (> 300 bp): there is a **gap in the centre** of the amplicon that is covered by neither R1 nor R2 (Figure 12).

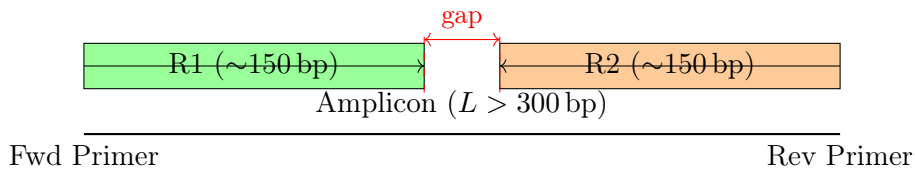


Figure 12: For amplicons longer than twice the read length (>300 bp with 150 bp reads), R1 and R2 do not meet, leaving an unsequenced gap in the centre of the amplicon.

2. No read-overlap means no overlap-based error correction. When paired reads overlap, the same genomic position is sequenced twice (once by R1, once by R2), enabling correction of sequencing errors by comparing the two observations. Since our amplicons are mostly longer than the read length, the reads do not overlap and this error-correction benefit is lost. As a consequence, a single-base mismatch in the non-overlapping region cannot be confirmed by the mate read, which **reduces our confidence in low-frequency variant calls** and makes quality-based filtering more important (see Section 8).

3. Adapter trimming is important for the minority of short fragments. Because most reads stop inside the amplified region, the majority never reach the far adapter. The ~ 6 – 11% adapter-containing reads originate from shorter amplicons or degraded fragments where the read length exceeds the insert size; these *must* be trimmed before alignment or the adapter sequence will cause mapping failures. For the remaining ~ 89 – 94% of reads, trimming only removes any low-quality 3' bases.

This also has an implication for **coverage uniformity** (Section 7): the ends of each amplicon (where both R1 and R2 contribute) will have higher coverage than the centre (reached by only one read direction), producing a characteristic *double-peak* coverage profile across long amplicons.

F Software and Tools

- **samtools** v1.19.2 — BAM handling, flagstat, idxstats, depth
- **FastQC** v0.12.1 — raw and trimmed read quality control
- **Cutadapt** v4.6 — adaptor contamination quantification and trimming
- **BLAST+** — primer localisation against hg19
- **bedtools** v2.31.1 — genomic interval operations and coverage
- **Mutect2** (GATK 4.5) — somatic SNV and indel calling
- **VarDictJava** v1.8.3 — somatic variant calling (amplicon-optimised)
- **SnpEff** v5.2 — variant functional annotation (hg19 database)
- **Snakemake** v7.32.4 — workflow management
- **Python** / **R** — data analysis and visualisation

References

- [1] Simon Andrews. FastQC: A quality control tool for high throughput sequence data. Babraham Bioinformatics, <http://www.bioinformatics.babraham.ac.uk/projects/fastqc/>, 2010.
- [2] David Benjamin, Takuto Sato, Kristian Cibulskis, Gad Getz, Chip Stewart, and Lee Lichtenstein. Calling somatic SNVs and indels with Mutect2. *bioRxiv*, 2019. Preprint.
- [3] Pablo Cingolani, Adrian Platts, Le Lily Wang, Melissa Coon, Tung Nguyen, Luan Wang, Susan J. Land, Xiangyi Lu, and Douglas M. Ruden. A program for annotating and predicting the effects of single nucleotide polymorphisms, SnpEff: SNPs in the genome of *Drosophila melanogaster* strain w1118; iso-2; iso-3. *Fly*, 6(2):80–92, 2012.
- [4] Sara Goodwin, John D. McPherson, and W. Richard McCombie. Coming of age: ten years of next-generation sequencing technologies. *Nature Reviews Genetics*, 17(6):333–351, 2016.
- [5] Illumina, Inc. *MiSeq System Guide*. Illumina, Inc., San Diego, CA, USA, 2020. Document no. 15027617 v03.
- [6] Lawrence J. Jennings, Maria E. Arcila, Christopher Corless, Suzanne Kamel-Reid, Ira M. Lubin, John Pfeifer, Robyn L. Temple-Smolkin, Karl V. Voelkerding, and Marina N. Nikiforova. Guidelines for validation of next-generation sequencing-based oncology panels: A joint consensus recommendation of the Association for Molecular Pathology and College of American Pathologists. *The Journal of Molecular Diagnostics*, 19(3):341–365, 2017.

- [7] Zhongwu Lai, Aleksandra Markovets, Miika Ahdesmaki, Brad Chapman, Oliver Hofmann, Robert McEwen, Jeffrey Johnson, Brian Dougherty, J. Carl Barrett, and Jonathan R. Dry. VarDict: a novel and versatile variant caller for next-generation sequencing in cancer research. *Nucleic Acids Research*, 44(11):e108, 2016.
- [8] Melissa J. Landrum, Jennifer M. Lee, Mark Benson, Garth R. Brown, Chen Chao, Shanmuga Chitipiralla, Baoshan Gu, Jennifer Hart, Douglas Hoffman, Wonhee Jang, Karen Karapetyan, Kenneth Katz, Chunlei Liu, Zenith Maddipatla, Adriana Malheiro, Kurt McDaniel, Michael Ovetsky, George Riley, George Zhou, J. Bradley Holmes, Brandi L. Kattman, and Donna R. Maglott. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018.
- [9] Marilyn M. Li, Michael Datto, Eric J. Duncavage, Shashikant Kulkarni, Neal I. Lindeman, Somak Roy, Apostolia M. Tsimberidou, Cindy L. Vnencak-Jones, Daynna J. Wolff, Anas Younes, and Marina N. Nikiforova. Standards and guidelines for the interpretation and reporting of sequence variants in cancer: A joint consensus recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. *The Journal of Molecular Diagnostics*, 19(1):4–23, 2017.
- [10] Thomas J. Lynch, Daphne W. Bell, Raffaella Sordella, Sarada Gurubhagavatula, Ross A. Okimoto, Brian W. Brannigan, Patricia L. Harris, Sara M. Haserlat, Jeffrey G. Supko, Frank G. Haluska, David N. Louis, David C. Christiani, Jeff Settleman, and Daniel A. Haber. Activating mutations in the epidermal growth factor receptor underlying responsiveness of non-small-cell lung cancer to gefitinib. *New England Journal of Medicine*, 350(21):2129–2139, 2004.
- [11] Marcel Martin. Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet.journal*, 17(1):10–12, 2011.
- [12] Tony S Mok, Yi-Long Wu, Myung-Ju Ahn, Marina C Garassino, Hye Ryun Kim, Suresh S Ramalingam, Frances A Shepherd, Yi He, Hiroaki Akamatsu, Willemijn S M E Theelen, et al. Osimertinib or platinum–pemetrexed in EGFR T790M-positive lung cancer. *New England Journal of Medicine*, 376(7):629–640, 2017.
- [13] Aaron R. Quinlan and Ira M. Hall. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, 26(6):841–842, 2010.
- [14] Jean-Charles Soria, Yuichiro Ohe, Johan Vansteenkiste, Thanyanan Reungwetwatana, Busyamas Chewaskulyong, Keunchil Lee, Arunee Dechaphunkul, Fumio Imamura, Naoyuki Nogami, Takayasu Kurata, et al. Osimertinib in untreated EGFR-mutated advanced non-small-cell lung cancer. *New England Journal of Medicine*, 378(2):113–125, 2018.
- [15] John G. Tate, Sally Bamford, Harry C. Jubb, Zbyslaw Sondka, David M. Beare, Nidhi Bindal, Harry Boutselakis, Charlotte G. Cole, Celestino Creatore, Elisabeth Dawson, Peter Fish, Bhavana Harsha, Charlie Hathaway, Steve Jupe, Chai Yin Kok, Kate Noble, Laura Ponting, Christopher C. Ramsden, Charlotte E. Rece, Andrew Reid, Ruth Shepherd, Iara Shih, Ann Tivey, Michael J. Wilkinson, Sheryl Ward, and Simon A. Forbes. COSMIC: the catalogue of somatic mutations in cancer. *Nucleic Acids Research*, 47(D1):D941–D947, 2019.

- [16] Yosef Yarden and Mark X. Sliwkowski. Untangling the ErbB signalling network. *Nature Reviews Molecular Cell Biology*, 2(2):127–137, 2001.